Predicting Formal Protest Petitions Against Housing Development: Evidence from Austin, Texas

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April 6, 2026

Abstract

This thesis asks whether the filing of a valid protest petition against a housing development proposal in Austin, Texas can be predicted before the project enters formal public hearings, using only information available at the time of filing. The primary outcome is an operationalized petition-based measure intended to capture whether a case met the study-period petition threshold in reconstructed public records. Working from an analytic sample of 7,074 discretionary zoning cases filed between 2007 and 2024, the formal protest petition model is evaluated under out-of-sample conditions: out-of-distribution bootstrap evaluation at the filing-date horizon yields a PR-AUC of 0.538 (95% CI: [0.3349, 0.7999]), with an Expected Calibration Error of 0.108 after Isotonic Regression—above the 0.05 threshold required for practical application. In-distribution 5-fold cross-validation provides an upper bound PR-AUC of 0.712 with a top-decile lift of 9.01× over the baseline petition rate. Filing-date administrative features contain meaningful signal for formal protest petition occurrence, but that signal is better suited to relative ranking than to forecasting that remains calibrated across policy periods. The false negative rate gap across council districts is 7.69%, but given the sparsity of positive cases within individual subgroups, this figure likely reflects insufficient statistical power rather than genuine equalization of error rates. Supporting analyses include a regression discontinuity design at the 20% petition threshold, which estimates a +52.1-day increase in council processing time (SE = 2.53, 95% CI: [47.09, 57.08]), and a difference-in-differences estimator for the 2022 council electoral transition, which identifies a -0.546 shift in petition filing rates in districts that changed political representation ($p = 0.041$). Semi-structured interviews with planning professionals contextualized these findings, highlighting the pre-2022 declining practical influence of the petition mechanism and the two distinct motivations driving public testimony. This thesis demonstrates that while administrative features contain strong predictive signal, the complex institutional realities of the zoning process are fundamentally difficult to capture through static model-based probabilities.

Acknowledgments

I would like to express my deepest gratitude to my thesis advisor, Dory Kornfeld, for her guidance, support, and expertise throughout this research. Her insights into urban planning and policy were invaluable in shaping this work.

I am profoundly grateful to my thesis readers for their time, expertise, and thoughtful feedback: Matthew Bauer, Adjunct Associate Professor; Josh Begley, Adjunct Assistant Professor; James Piacentini, Adjunct Assistant Professor; Stijn Van Nieuwerburgh, Earle W. Kazis and Benjamin Schore Professor of Real Estate and Co-Director of the Paul Milstein Center for Real Estate at Columbia Business School; and Adam Vosburgh, Associate Director and Adjunct Assistant Professor. Their interdisciplinary perspectives spanning computational design, real estate finance, and urban planning enriched this research immeasurably.

I would like to thank Professor Weiping Wu, Vice Provost for Academic Programs and Director of the Urban Planning Program at Columbia University, for her leadership of the program and support of my academic journey.

I am grateful to the Columbia University Graduate School of Architecture, Planning and Preservation (GSAPP) for providing the academic environment and resources that made this research possible.

Finally, I thank my colleagues, peers, and the broader Columbia community for their intellectual engagement, encouragement, and support throughout this journey.

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1 Introduction

As the United States confronts a persistent housing affordability crisis, understanding local barriers to new housing supply remains an important empirical challenge. Traditional accounts emphasize the role of municipal zoning laws, but a substantial share of delay and blockage occurs through discretionary political processes—public hearings, protest petitions, and council negotiations—that collectively slow or block individual projects. This thesis uses Austin, Texas as a case study. Between 2010 and 2024, rapid employment growth drove severe housing supply constraints, inflating property values and generating sustained conflict between neighborhood preservation and regional development goals.

Against this backdrop, the central research question is whether a valid protest petition filing can be predicted before formal public hearings begin.

The outcome under study is specific. Under the pre-2025 framework of Texas Local Government Code § 211.006 [29], adjacent property owners may file a protest petition; if signatures representing 20% of the relevant adjoining land area were collected and the municipal clerk certified the petition as valid, the petition historically required a three-fourths supermajority for City Council approval. (Effective September 1, 2025, House Bill 24 repealed and replaced this mechanism with a differentiated, multi-tier system under § 211.0061.) This is one institutionalized form of neighborhood opposition—the most consequential and best-observed form in the Austin regulatory context—not a prediction of opposition in all its forms. The thesis narrows its claim accordingly: the primary contribution is an assessment of whether administrative, spatial, and economic features observable at the time of a zoning application are sufficient to anticipate this specific institutional event.

To answer this question, the research constructs a sequential predictive methodology organized around four stages:

- **Stage A—Development Occurrence:** Where housing development proposals are likely to emerge, modeled as a discrete-time site hazard across 282,000 candidate parcels.
- **Stage B—Project Type and Scale:** Conditional on a development event, the expected project form and unit count.
- **Stage C—Formal Protest Petition:** Conditional on a project entering the formal public process, the probability that a valid protest petition is filed. *This is the core of the thesis.*
- **Stage D—Political Outcome:** Conditional on project characteristics and opposition status, the likelihood of pre-council withdrawal or final disposition.

Stages A, B, and D provide context for Stage C. Stage A supplies inverse-probability weights to adjust for spatial selection into the analytic sample. Stage B characterizes the heterogeneous institutional responses associated with different project types. Stage D documents the downstream consequences of petition filing, including high rates of pre-council withdrawal among opposed cases. The predictive evaluation, however, centers on Stage C: whether the model can discriminate petition risk at the filing-date horizon and whether it does so with sufficient calibration and equity to support high-confidence administrative use.

The thesis also includes two exploratory quasi-experimental analyses of downstream institutional dynamics. A regression discontinuity design at the 20% petition threshold estimates the processing-time penalty imposed by crossing the statutory margin. A difference-in-differences estimator exploits the geographic heterogeneity of the November 2022 council elections, in which specific districts replaced preservationist incumbents with pro-housing representatives, to test whether political turnover reduced petition filing rates in treated districts.

Finally, semi-structured interviews with planning professionals, conducted under IRB Protocol ACYY0820, informed the interpretation of quantitative findings by surfacing institutional dynamics—informal negotiations, strategic petition timing, neighborhood network mobilization—that administrative data cannot directly observe. The interviews do not validate the model; they contextualize findings within professional planning practice and surface governance concerns about potential misuse of predictive outputs.

2 Literature Review

This literature review synthesizes scholarship across three core themes to directly motivate the thesis's methodological architecture. It first establishes the political economy of homeowner opposition and participatory distortion. Second, it isolates the specific institutional mechanism of the protest petition. Finally, it provides the intellectual justification for treating zoning conflict as primarily a predictive policy problem while acknowledging the inherent evaluative constraints required for administrative use of predictive models in spatial planning.

2.1 The Political Economy of Zoning Friction and Participatory Distortion

The foundational context for this research rests on the consensus that land-use regulation functions as a binding constraint on housing supply, driving a wedge between fundamental construction costs and market prices [14, 13]. While macro-economic literature establishes the downstream effects of this regulatory tax—including suppressed economic growth and diminished housing affordability—urban planning scholarship points to the local political economy to explain why such constraints persist despite citywide needs [15].

The theoretical centerpiece for understanding this localized resistance is Fischel's [12] homevoter hypothesis. Fischel posits that homeowners engage in local politics primarily to protect both the use value and exchange value of their homes, rationally utilizing zoning restrictions to preserve neighborhood exclusivity and mitigate perceived financial risks. However, recent empirical work has demonstrated that this protective impulse translates into a highly asymmetric political arena. Einstein, Palmer, and Glick [11] document the phenomenon of "participatory distortion," demonstrating that the participants who dominate planning and zoning board meetings are overwhelmingly older, disproportionately white, longtime homeowners who are united in opposing new housing. These "neighborhood defenders" possess the time, resources, and localized knowledge to effectively block individual projects. Furthermore, Hankinson [17] complicates the pure asset-protection motive by revealing that renters in high-cost cities exhibit spatial NIMBYism on par with homeowners due to localized displacement anxiety, cementing the reality that spatial proximity to development triggers mobilization regardless of tenure.

2.2 Discretionary Review and Institutional Vetoes

Participatory distortion matters for housing supply when it operates through institutions that can delay or block projects. Manville and Monkkonen [22] highlight that discretionary review processes—where development relies on site-specific municipal approvals rather than by-right administrative clearance—serve as the primary vehicle through which local opposition exerts leverage. In a discretionary regime, citywide housing goals are repeatedly displaced by parcel-level political conflict [19].

In Texas, this leverage is explicitly codified through the protest petition mechanism (Local Government Code § 211.006). The statute legally empowers adjacent property owners to force supermajority city council voting requirements if owners representing 20% of the *area of lots* immediately adjoining the proposed change and extending 200 feet from that area sign a valid petition.[1] This mechanism empowers localized opposition, transforming protest into a formal institutional hurdle. Cellini, Ferreira, and Rothstein [6], in their analysis of California school bond referenda, demonstrate how supermajority thresholds inherently distort political outcomes to favor the status quo.

In Austin, the protest petition creates institutional friction for land use: cases exceeding the measured petition threshold appear to face additional voting requirements and longer processing times. Recent empirical research by the Mercatus Center at George Mason University confirms the severity and asymmetry of this mechanism specifically within the Austin political geography. Their analysis demonstrates that valid petitions are more common for rezonings that increase residential density: historically, 25% of rezonings to multifamily use in Austin faced a valid petition, compared to only 5% for commercial rezonings.

Table 1: Valid Petition Incidence by Proposed Land Use in Austin (Mercatus Center)

Proposed Zoning Category	**Rezonings Facing Valid Petition (%)**
Multifamily Residential	25%
Commercial / Non-Residential	5%

[1] Note that § 211.006(d) and (f) were repealed effective September 1, 2025 by HB 24 (89th Texas Legislature). The operative provisions now reside in new § 211.0061, which establishes a three-tier protest system.

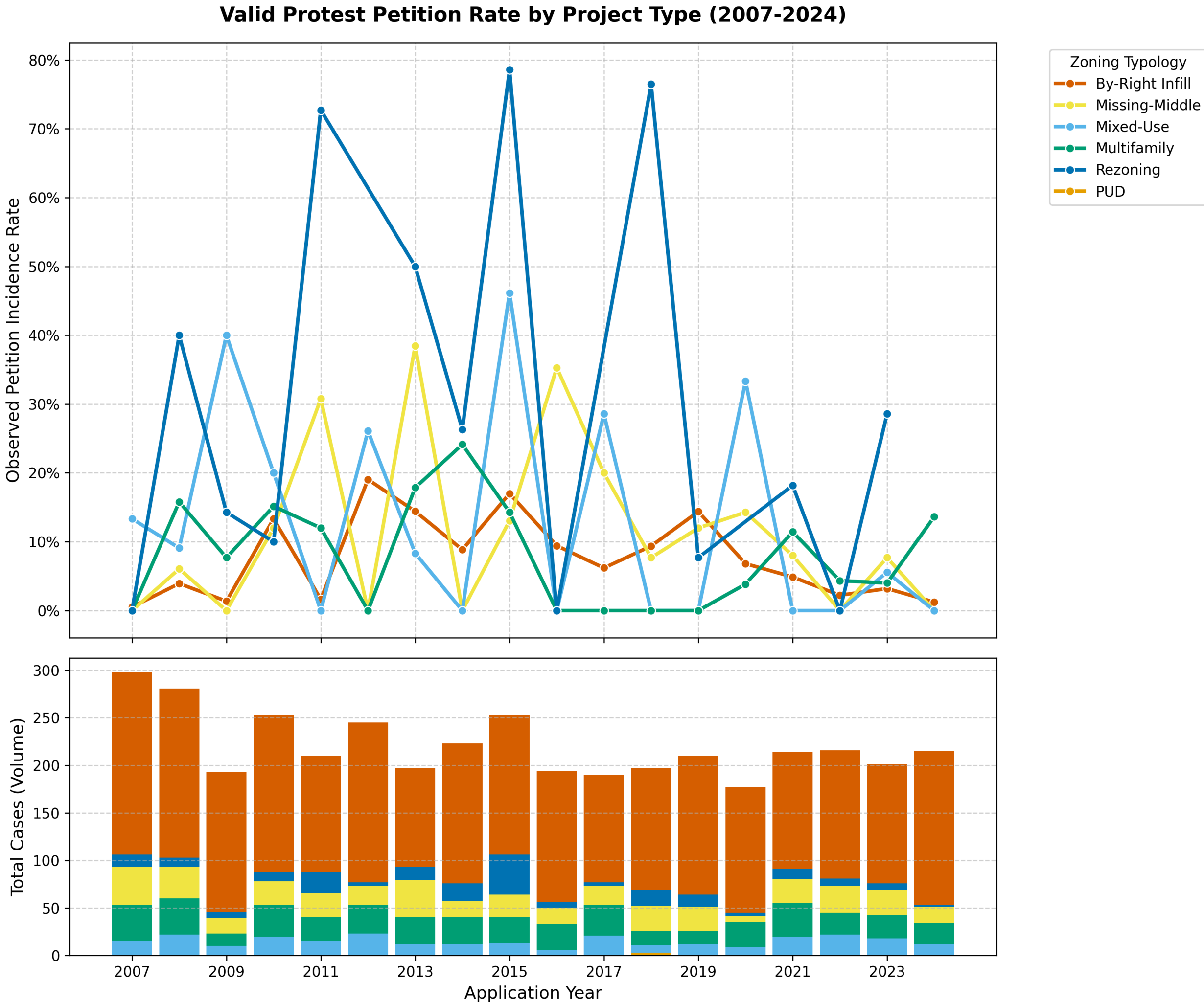


Figure 1: Annual petition filing rates by project type, 2007–2024. Consistent with the Mercatus Center pattern (Table 1), this timeline tracks the observed historical petition frequency within each development category over the study period. While high-density or specialized use categories historically experience greater petition volatility, missing-middle and standard residential rezonings—which comprise the vast majority of all discretionary cases—maintain a consistently low baseline filing rate throughout the dataset.

Because the process is fundamentally asymmetric—empowering opponents of development through a fixed statutory buffer geometry—it creates a bias toward the status quo, dragging contested housing applications into months of procedural delays. The Texas Legislature structurally altered this framework in 2025; under House Bill 24, the state established new § 211.0061, replacing the uniform 20% trigger with differentiated standards based on project type and requiring a two-thirds rather than three-fourths council vote for certain residential rezonings (while largely preserving the prior framework for nonresidential cases). Consequently, predicting whether a qualifying formal protest petition will be filed against an application forms the core empirical challenge of this thesis.

2.3 The Prediction Policy Problem in Spatial Planning

Traditional econometric scholarship in urban planning almost exclusively evaluates the causal effect of zoning—for instance, whether upzoning increases land values or whether a protest petition causes delay. However, addressing the localized gridlock of discretionary review requires anticipating *where* opposition will emerge before political capital is expended. This intellectual justification builds on Shmueli's [28] foundational distinction between explanatory modeling (testing causal hypotheses) and predictive modeling (generating accurate forecasts of new observations), and on Mullainathan and Spiess [23], who argue that machine learning solves a fundamentally different problem than traditional econometrics: prediction ($\hat{y}$) rather than parameter estimation ($\hat{\beta}$).

Kleinberg, Ludwig, Mullainathan, and Obermeyer [21] formalize this distinction by defining the "predictive policy question," cases where optimal policy decisions depend not on understanding the causal mechanism of an outcome, but on accurately forecasting its occurrence. Predicting which zoning cases will meet the petition threshold can be framed as a predictive policy question. Planners and policymakers need to know which projects will trigger supermajority voting requirements to efficiently allocate political capital, engage in proactive community mediation, or design comprehensive reforms that bypass parcel-level friction [2]. By applying algorithmic forecasting to administrative municipal data, this research transitions the study of NIMBYism from a retrospective causal analysis into a proactive predictive model, testing whether administrative spatial characteristics carry sufficient signal to anticipate formal protest petitions in formal proceedings weeks or months before a public hearing takes place.

2.4 Evaluative Constraints on Predictive Governance

While transitioning to a predictive methodology offers potential analytical value for municipal planning, it introduces serious ethical risks. Barocas and Selbst [3] demonstrate that predictive algorithms inevitably inherit the prejudices embedded in their training data. Because historical zoning data in American cities is inextricably linked to legacies of redlining, racial covenants, and exclusionary zoning [31], the underlying spatial features, such as lot geometry, single-family zoning, and historic property values, function as proxies for race and class. If an algorithm learns that wealthy, historically white neighborhoods successfully file protest petitions, using that prediction to guide development decisions risks encoding a feedback loop that shields those neighborhoods from densification while concentrating development pressure on politically vulnerable communities [27].

Consequently, the viability of algorithmic prediction in spatial planning is constrained by mathematical limits on fairness. Chouldechova [8] proves that when base rates of an outcome differ across groups, no imperfect predictor can simultaneously satisfy both predictive parity and classification parity. Recognizing these fixed mathematical constraints, modern predictive models intended for public-sector use must be evaluated against explicit calibration and error-rate thresholds. Rather than attempting to maximize pure predictive lift, the administrative application of such models should consider explicit calibration accuracy and the equalization of False Negative Rates across geographic groups, ensuring that predictive efficiency does not come at the cost of democratic equity.

3 Data and Methods

The empirical foundation is a time-stamped spatial panel dataset that reconstructs what information was available at each historical prediction point. Every record anchors to an explicit as-of date, ensuring that backtests use only Census releases, appraisal rolls, and case documents published before the prediction date.

3.1 Analytical Panel Structure

The core analytics rely on the filing-date analytic dataset, which merges spatial and demographic variables into single case-level observations (with temporally matched covariates):

1. **Zoning Application Records:** Case ID, milestone dates (filing, notice, petition deadline, planning commission hearing, council reading, withdrawal), requested zoning changes, unit limits, target FAR, and disposition status.

2. **Spatial Geometries:** Parcel IDs, geometries, acreage, frontage, and 200ft buffers used to aggregate nearby properties (council districts and overlapping geometries).

3. **Parcel Appraisals:** Annually indexed TCAD appraisal variables for parcels used to compute variables within the 200-foot buffer: land value, structure age, homestead exemption rates, and owner-occupancy shares.

4. **Neighborhood Demographics:** Tract and block group ACS variables constrained to their historical 5-year release dates: rent burdens, tenure splits, demographic composition, and displacement vulnerability indicators.

5. **Institutional Calendar:** The 2022 council regime change; the HB 24 effective date (September 1, 2025).

Because municipal databases frequently overwrite preliminary timestamps with final disposition dates, precise milestone dates are reconstructed through two methods. First, explicit dates are parsed from scraped historical council agendas and PDF case files. Second, where explicit text is unavailable, dates are imputed by back-calculating from end-dates using the minimum procedural constraints mandated by Texas Local Government Code and Austin municipal zoning ordinances, such as statutory 11-day mail notice minimums.

Supporting tables log meeting events (staff and commission recommendations), speech comments (speaker stance and segmented text), petition records (validity and signature counts), and vote records (case-by-councilmember dynamics).

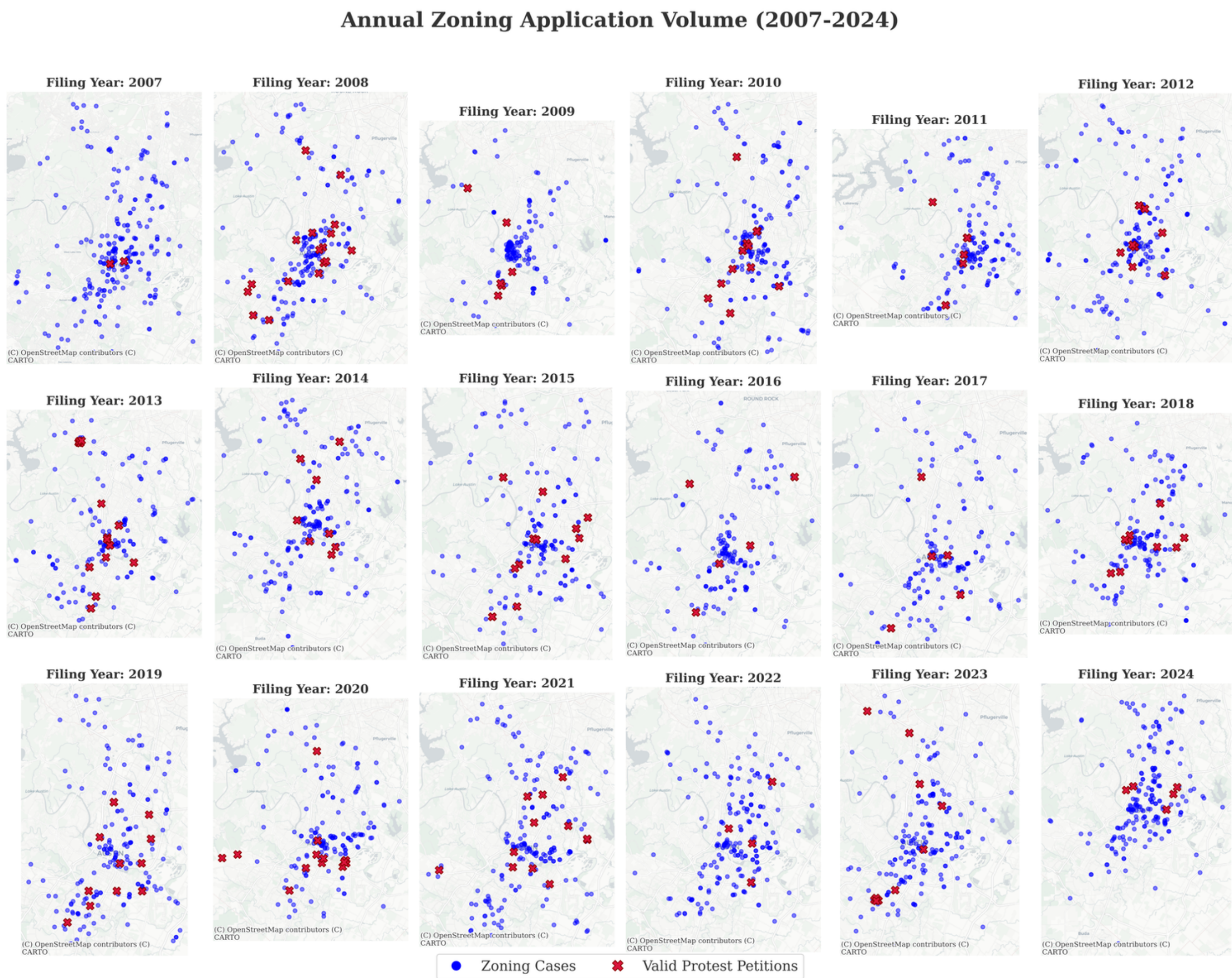


Figure 2: Spatial distribution of zoning cases across the Austin municipal core ($N = 7,074$).

Table 2: Annual Distribution of Discretionary Zoning Cases (2007—2024)

Year	Cases	Year	Cases	Year	Cases
2007	288	2013	366	2019	460
2008	270	2014	390	2020	362
2009	265	2015	454	2021	476
2010	297	2016	422	2022	436
2011	369	2017	483	2023	625
2012	335	2018	492	2024	284
Total (2007—2024): 7,074 cases					

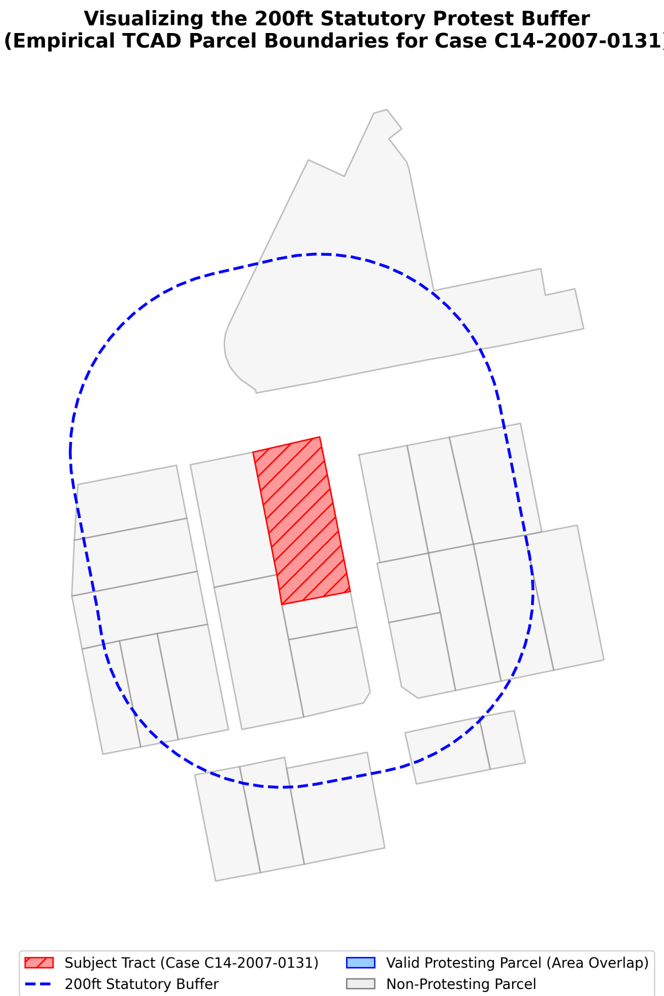


Figure 3: 200ft parcel buffer geometries used for TCAD appraisal aggregation and defining the petition buffer.

3.2 Case Universe and Sample Selection

The Austin Open Data Portal (data.austintexas.gov), accessed via the Socrata Open Data API (SODA), records all administrative zoning activity, including both substantive rezoning proposals and minor administrative variances, such as 5-foot setback adjustments. Because the petition mechanism under study—formal protest petitions under Texas LGC Chapter 211—does not apply to minor variances, and because extraterritorial jurisdiction (ETJ) properties lack the legal protest-petition mechanism, these cases are excluded.

Figure 4 illustrates the municipal zoning process, highlighting where the 20% protest petition threshold creates a potential supermajority requirement at City Council.

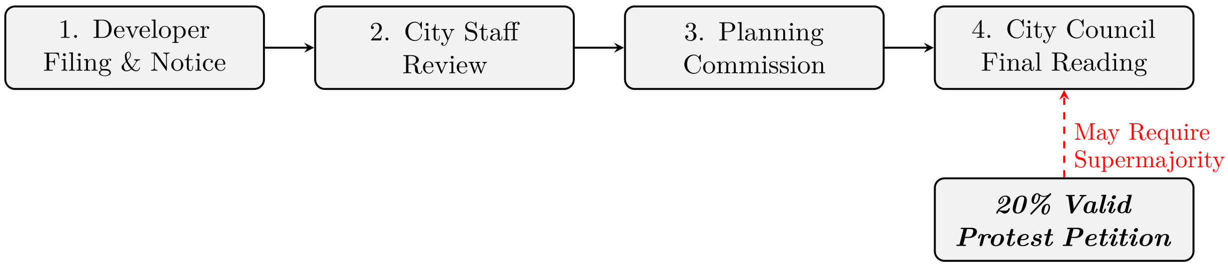


Figure 4: Austin municipal zoning process. The figure schematically shows where a qualifying protest petition could make a three-fourths council vote relevant under the study-period framework. A valid protest petition representing 20% of the area of lots immediately adjoining the proposed change would require, under the applicable legal framework, a supermajority vote of the City Council for approval, though recusal or absence of council members can alter the exact number of affirmative votes needed.

Table 3: Historical Panel Descriptive Statistics

	count	mean	std	min	Median	max
Gross Site Area (Acres)	534	4.92	30.66	0.00	0.00	669.83
Requested Height Delta (ft)	3994	10.5	34.9	-60	0	365
Requested FAR Delta	3994	0.43	0.95	-3.00	0.05	7.60
Requested Bldg Cov Delta (%)	3994	12.3	20.7	-55	5	80
Filing Year	7074	2008.6	10.11	1966	2009	2026
Organized Opposition (Binary)	7153	0.06	0.23	0	0	1

Table 4 reports the sample selection from raw administrative records to the analytic sample.

Table 4: Sample Selection from Raw Records to Analytic Sample

Selection Step	**Cases Remaining**
Raw Austin open data portal zoning records (2007 to 2024)	15,238
Restrict to discretionary cases (rezonings, PUDs, NPAs)	7,153
Exclude cases with unusable or missing initial filing-year information	7,074
Final Analytic Sample (Stages C/D)	**7,074**

The analytical samples vary by analytical goal. The primary predictive evaluation (Stage C) uses the full sample of 7,074 discretionary geographic cases. In contrast, the causal quasi-experiments are restricted sub-samples designed to isolate specific mechanisms. The 2022 Difference-in-Differences and the Event Study evaluations utilize $N = 518$ because they are restricted to active single-member districts immediately surrounding the 2022 election cycle. The Invariant Causal Prediction (ICP) analysis utilizes $N = 1,186$ multi-parcel geometric environments to represent cohesive spatial clusters. Finally, the institutional geographic overlay regressions evaluate policy flags aggregated at the council-term level across periods, resulting in $N = 20$ macro-observations.

Unlike approaches that require complete-case geometric intersections, this methodology uses CatBoost's native missing-value handling to build splits over absent variables, such as unmatched TCAD boundaries, without listwise deletion. This approach retains cases with partially missing covariates, reducing artificial attrition and the selection bias introduced by demanding full covariate coverage. Whether it also improves fairness metrics is a separate empirical question explored in Section 7.

3.3 Primary Outcome Definition

The primary predictive outcome (y_0) is whether a valid protest petition is filed under the 20% statutory threshold defined by Texas Local Government Code Chapter 211 and Austin LDC § 25-2-284. This is an operationalized outcome derived from petition filings and measured petition-share calculations recorded through public petition filings.

Secondary outcomes are analyzed separately:

- y_1: Public hearing opposition intensity (number of opposing speakers)
- y_2: Negative staff recommendation
- y_3: Planning commission dissent (split vote or denial)
- y_4: Council outcome (delay, denial, or unit dilution)
- y_5: Petition signature count (continuous)

All primary results report y_0. Sensitivity analyses replicate key findings using individual secondary outcomes. This separation is important because staff recommendations, commission votes, and council outcomes reflect different institutional mechanisms than neighborhood mobilization through petitions, and collapsing them into a single binary would obscure meaningful distinctions. Cases withdrawn before public notice are excluded, rather than coded as zero, because no opportunity for opposition existed. ETJ cases and routine variances are excluded because the formal petition mechanism does not apply to them.

Note on stage-specific samples: the Stage C filing-date model uses the full 7,074-case analytic sample regardless of whether a hearing transcript exists, since transcripts post-date the filing horizon and are not used as predictive features at that stage. Cases missing hearing transcripts or petition-database coverage are omitted only from later-horizon evaluations and secondary-outcome analyses where those records serve as inputs or outcome sources.

3.4 Feature Libraries: Policy Forecasting vs. Explanatory Research

The feature set is separated to prevent predictive models from leveraging sensitive attributes:

Policy-forecasting features include physical site characteristics (lot area, unit count changes, PUD/TOD flags), statutory eligibility flags (HB 24 status, owner-vs.-city-initiated), aggregated buffer economics (homestead shares, land-to-improvement ratios), broad demographic indicators (median income, renter share), historical petition rates within the council district, and macroeconomic variables (mortgage rates, vacancy rates). These elements are selected strictly based on their availability in public administrative and appraisal databases prior to the initial filing date, ensuring the algorithm's decisions replicate constraints faced by municipal planning staff in real time.

Explanatory research features are restricted to equity analysis. These include tract-level racial and ethnic composition, segregation indices, and probabilistic surname-based racial classification. These features are strictly partitioned from the predictive model to reduce the risk of encoding disparate impacts, though proxy correlation through permitted features remains a concern (see Section 7). By reserving these variables exclusively for post-hoc fairness evaluations, the architecture avoids generating predictions based on protected class attributes while maintaining the capability to measure intersectional disparities across geographic subgroups.

4 Modeling Strategy

4.1 Development Occurrence

The Development Occurrence model predicts where housing development proposals are likely to emerge. Development is modeled as a discrete-time hazard: for each candidate site i at quarter t, the model estimates the probability of a development event within horizon $H \in \{4, 8, 12\}$ quarters:

$$h^{dev}_{i,t,H} = P(\text{development event in next } H \text{ quarters} \mid X_{i,t}) \tag{1}$$

The unit of analysis is a candidate site-quarter, drawn from over 282,000 baseline Austin parcels aggregated to reflect contiguous configurations over time. The primary model architecture uses CatBoost discrete-time hazard estimation, regularized to isolate spatial and economic feasibility signals such as improvement-to-land ratios, accessibility, and zoning capacity. Table 5 defines the nested target outcomes.

Table 5: Stage A Target Definitions

Target	**Definition**	**Type**
A1	Any housing production event (rezoning, permit, demolition)	Binary
A2	Discretionary zoning application subject to public hearing	Binary
A3	Net-new housing unit deliveries	Continuous

Because the baseline class imbalance is extreme—across 5,089,896 site-quarter observations, the empirical probability of a development event at any single site-quarter is approximately 2.5×10^{-5}—performance must be evaluated by relative lift rather than raw accuracy. Table 6 reports this hazard model performance relative to the baseline probability.

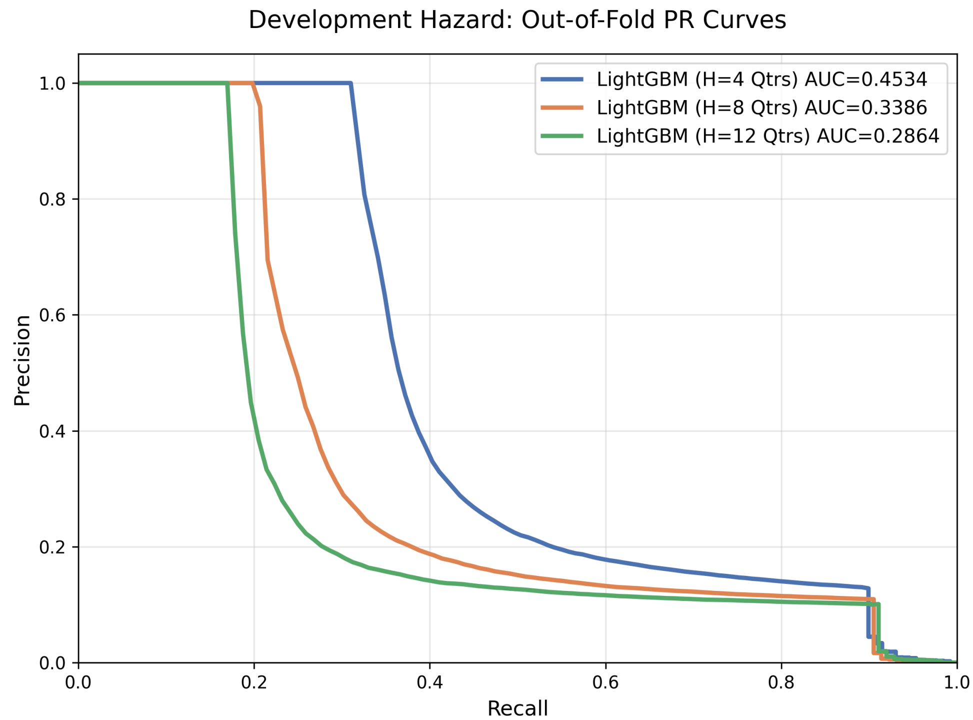


Figure 5: Precision–recall curves for the filing-date petition model across evaluation horizons. Note: Out-of-fold evaluations represent aggregate discrimination prior to restricting maps to the top decile of predicted risk.

Table 6: Development Occurrence Hazard Model Performance

Horizon	**PR-AUC**	**Algorithm**	**Lift over Baseline**[†]
4-Quarter	0.1917	LightGBM	$\sim$7,600$\times$
8-Quarter	0.1201	LightGBM	$\sim$4,800$\times$
12-Quarter	0.1127	LightGBM	$\sim$4,500$\times$

[†]Lift defined as PR-AUC $\div$ base hazard probability ($\approx 2.5 \times 10^{-5}$). This is not a top-decile concentration lift; it reflects the extreme class imbalance of the site-quarter panel.

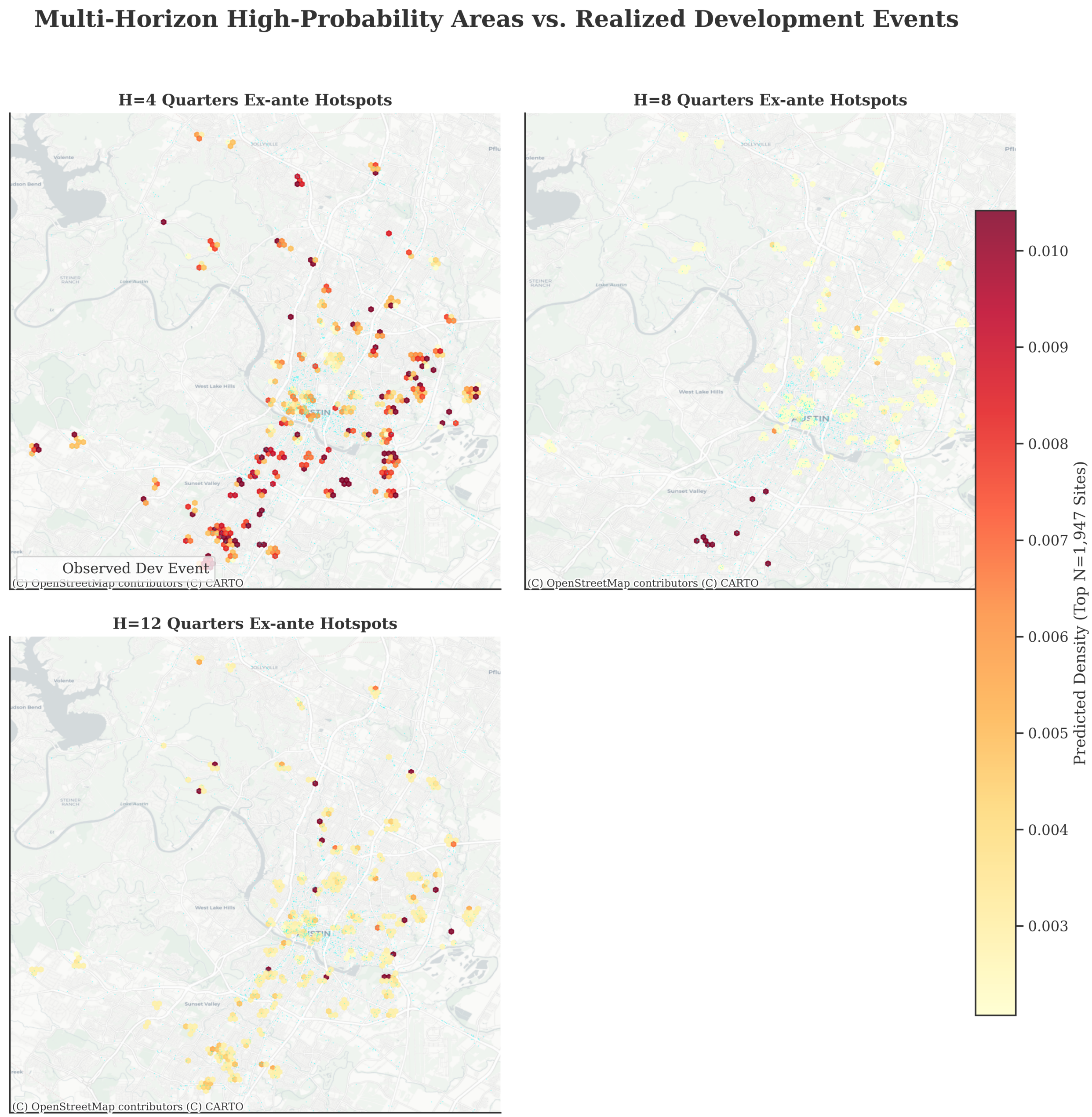


Figure 6: Ex-ante predicted areas of high predicted incidence density vs. realized development events. To define areas of high predicted incidence against the very low base rate, the map filters exclusively for the top decile (Top 10%) of parcels with the highest predicted development hazard, while displaying all realized zoning cases over the period. This disparity contributes to the visual imbalance where the number of realized cases outnumbers the parcels retained after restricting to the top decile of predicted risk. Hexbins requiring a minimum point-density count are used to visually suppress isolated sites and isolate clusters.

4.2 Project Type and Scale

Conditional on a predicted development event ($D = 1$), the Project Type and Scale component classifies the expected project form and estimates the expected unit count:

$$E(\text{net new units}_{i,t,H} \mid D = 1, X_{i,t}) \tag{2}$$

This stage is critical because a six-unit by-right infill project generates fundamentally different institutional responses and petition likelihoods compared to a 300-unit Planned Unit Development (PUD). The taxonomy classifies proposals into six categories defined by approval pathway and building form: PUD/Large Negotiated, Discretionary Rezoning, By-Right Infill, Missing-Middle, Mixed-Use, and Multifamily. By segmenting proposals by type, the framework isolates these heterogeneous risks. Table 7 reports classification performance.

Table 7: Project Type Classification Performance. Macro-F1 scores calculate the unweighted simple average of F1 scores computed individually for each of the six project types (C). Defined as $Macro\text{-}F_1 = \frac{1}{|C|}\sum_{c \in C} F_{1,c}$, macro-F1 penalizes poor performance on minority classes (e.g., PUDs) rather than rewarding accuracy only on more common classes.

Component	**LightGBM**	**Random Forest**	**CatBoost**	**Logistic (L2)**
Macro-F1 Baseline	0.748	0.739	0.548	0.177
Class-Level Typology Performance (F1)				
↪ PUD / Large Negotiated	**1.000**	**1.000**	**1.000**	0.097
↪ Discretionary Rezoning	0.850	**0.863**	0.724	0.455
↪ By-Right Infill	0.751	**0.757**	0.704	0.045
↪ Missing-Middle	**0.519**	0.507	0.392	0.257
↪ Mixed-Use	**0.789**	0.718	0.302	0.066
↪ Multifamily	0.579	**0.590**	0.165	0.140

4.3 Formal Protest Petition

The Formal Protest Petition forecast is the core of the thesis: conditional on a project entering the formal public process, predicting the probability of a valid protest petition ($y_0 = 1$):

$$P(y_0 = 1 \mid D = 1, X_{i,t}, Z) \tag{3}$$

At the initial filing date horizon, the predictive feature space ($X_{i,t}$) is restricted to information observable by planners at the time of filing, including site geometry, requested zoning changes, buffer demographics, and historical petition rates. To adjust for selection on observables—given that the analytic sample naturally contains only neighborhoods that actually receive development proposals—the model applies inverse-probability weighting derived from the Development Occurrence hazard (Z). While this weighting corrects for spatial baseline probabilities, it remains inherently limited because it cannot completely resolve unobserved political or social factors that simultaneously influence both proposal location and opposition likelihood. Precision-recall curves comparing the CatBoost, Random Forest, ElasticNet Logistic, and V-REx models are reported in Figure 7.

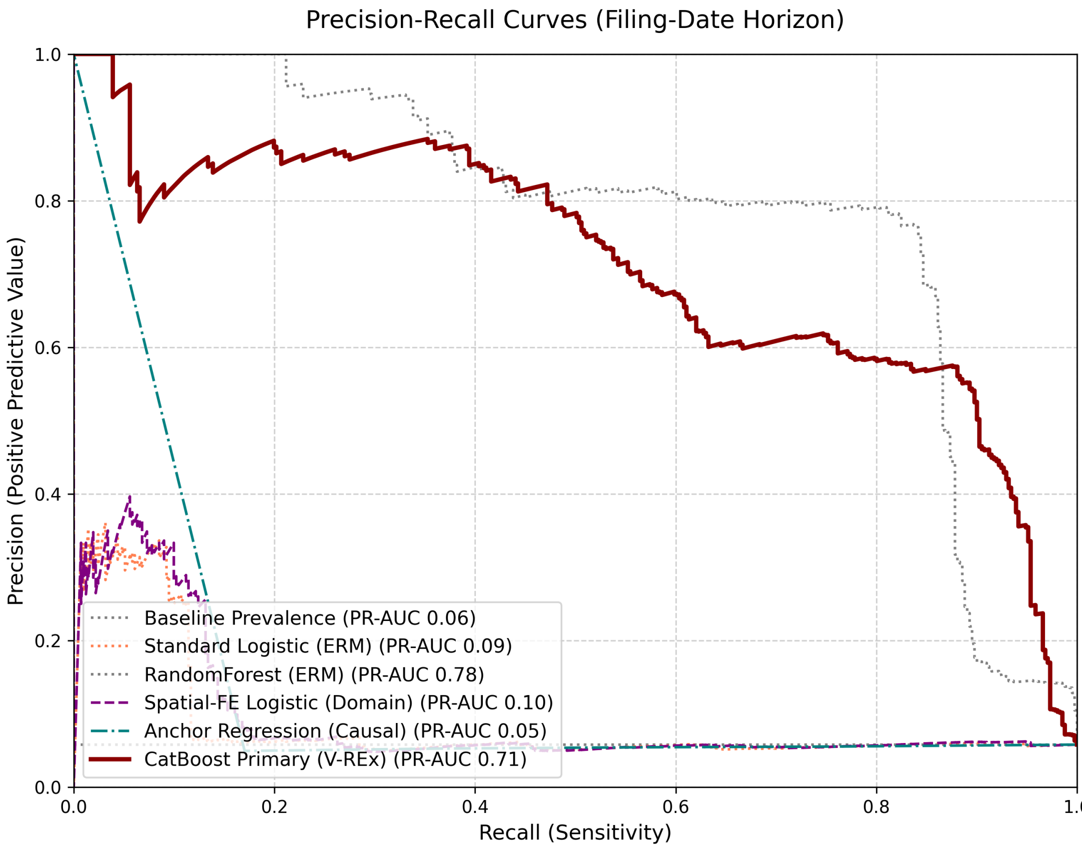


Figure 7: Precision–recall curves for the filing-date petition model: CatBoost, Random Forest, ElasticNet Logistic, and V-REx.

Table 8: Precision-Recall Area Under Curve (PR-AUC) Evaluation

Model	**PR-AUC**	**Lift over Baseline**
CatBoost Primary (V-REx)	0.7118	12.25×
RandomForest (ERM)	0.7836	13.49×
Anchor Regression (Causal)	0.0543	0.93×
Spatial-FE Logistic (Domain)	0.0951	1.64×
Standard Logistic (ERM)	0.0873	1.50×

A comprehensive benchmark comparison of alternative models (Random Forest, LightGBM [20], PyTorch [24] TabNet [1], with and without SMOTE [7] resampling) is provided in Table **??**. CatBoost [26] was retained for the primary architecture because of its native handling of categorical variables without manual encoding and its more stable calibration across evaluation settings, though the performance differences across models are modest.

Table 9: Development-proposal and petition-filing models evaluated at filing: Raw vs. calibrated candidate models

Horizon	Model	PR-AUC	ROC-AUC	ECE (Raw)	ECE (Cal)	ACE (Cal)
H0 (Filing)	Logistic Regression (L2) (Base)	0.064	0.731	0.586	**0.149**	0.022
H0 (Filing)	SMOTE + Logistic Regression (L2)	0.060	0.751	0.552	0.417	0.280
H0 (Filing)	Random Forest (Base)	0.422	0.967	0.298	0.399	**0.004**
H0 (Filing)	SMOTE + Random Forest	0.418	0.966	0.356	0.219	0.036
H0 (Filing)	LightGBM (Base)	0.489	0.962	0.375	0.225	0.012
H0 (Filing)	SMOTE + LightGBM	0.416	0.933	0.346	0.276	0.013
H0 (Filing)	CatBoost (Base)	**0.492**	0.962	**0.251**	0.285	0.009
H0 (Filing)	SMOTE + CatBoost	0.459	0.961	0.301	0.312	0.023
H0 (Filing)	TabNet	0.410	**0.972**	0.268	0.250	0.023
H3 (Pre-Council)	Logistic Regression (L2) (Base)	0.419	0.926	0.479	0.264	0.032
H3 (Pre-Council)	SMOTE + Logistic Regression (L2)	0.412	0.926	0.479	0.423	0.209
H3 (Pre-Council)	Random Forest (Base)	0.500	**0.963**	0.222	**0.127**	**0.010**
H3 (Pre-Council)	SMOTE + Random Forest	0.536	0.946	**0.171**	0.191	0.031
H3 (Pre-Council)	LightGBM (Base)	**0.558**	0.932	0.384	0.205	0.011
H3 (Pre-Council)	SMOTE + LightGBM	0.479	0.914	0.410	0.346	0.036
H3 (Pre-Council)	CatBoost (Base)	0.492	0.926	0.404	0.241	0.012
H3 (Pre-Council)	SMOTE + CatBoost	0.482	0.946	0.353	0.255	0.038
H3 (Pre-Council)	TabNet	0.456	0.944	0.271	0.235	0.017

4.4 Political Outcome

Conditional on both a development proposal and the presence of a valid petition, Stage D estimates the probability that a case survives to a final council reading rather than being withdrawn, stalled, or canceled before a vote occurs. Because virtually all cases that reach a final vote are approved (see Section 6), terminal council approval is near-deterministic and uninformative as a predictive target. Modeling pre-council attrition instead captures the institutional friction through which opposition most often affects outcomes.

Table 10: Pre-Council Attrition Forecast (Opposed Cases Only)

Model Model	**Log Loss**	**Accuracy**
LightGBM	**0.0255**	**0.9976**
CatBoost	0.1752	0.9440
Random Forest	0.2294	0.9854
Penalized Logistic (L2)	0.5951	0.3577

Accuracy and log loss are reported for this stage because the outcome (pre-council attrition/withdrawal) is highly imbalanced among opposed cases (77.6% attrition rate), making macro PR-AUC less informative as a single metric than threshold-independent probability scoring (Log Loss).

4.5 Expected Contested Units

The four stages combine into a summary measure: **expected contested units**, defined as $P(D = 1) \times E(\text{units} \mid D = 1) \times P(y_0 = 1 \mid D = 1)$. This identifies locations where the city is most likely to encounter opposition to housing development. However, the multiplicative structure compounds uncertainty from each stage: errors in the development hazard, the unit-count forecast, and the petition risk model propagate jointly, so the resulting map should be interpreted as a relative ranking rather than a reliable absolute probability. Figure 8 maps this composite measure.

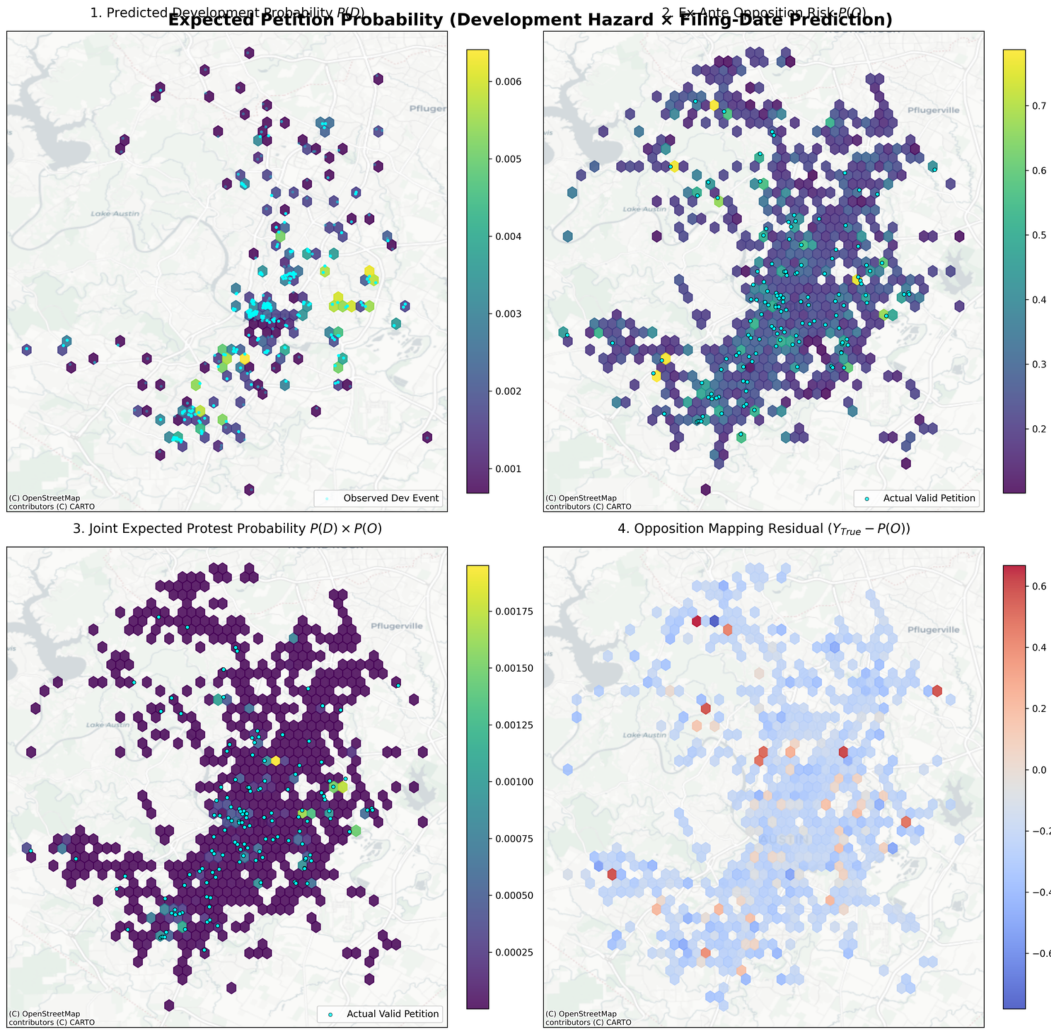


Figure 8: Expected units in projects facing valid protest petitions (Stage A: H=4 Quarter Hazard $\times$ Stage C: filing-date prediction). Cyan dots in Panel A represent observed historical development zoning cases. Panels B and C highlight actual valid petition events mapped over the composite risk and prediction variance.

4.6 Model Evaluation Strategy

Precision-Recall AUC (PR-AUC) is the primary discrimination metric, given the substantial class imbalance of the outcome. Following standard methodological practices in rare-event spatial forecasting, traditional Receiver Operating Characteristic (AUROC) metrics are omitted entirely, as the large number of true negatives artificially inflates the curve. PR-AUC explicitly isolates the minority class, forcing the architecture to balance recall capabilities against the precision cost of false alarms.

Because the baseline opposition frequency is highly skewed, raw predicted probabilities from gradient boosting models suffer from persistent overconfidence. To address this, the architecture applies out-of-fold Isotonic Regression, which fits a non-decreasing, non-parametric monotonic function mapping predicted probabilities to empirical outcome frequencies, enforcing calibration within the evaluation loop. Alongside standard Expected Calibration Error (ECE), the evaluation also reports the Brier Score and Adaptive Calibration Error (ACE = 0.1113). At a classification threshold of $P \geq 0.50$, the model achieves precision of 80.0% and recall of 93.2%, flagging 479 of 7,074 cases. The Brier score is 0.0156.

Strictly proper scoring rules like the Brier Score [4] and Adaptive Calibration (which evaluates equal-mass rather than equal-width bins) are useful complementary calibration measures; standard ECE can be unstable under severe class imbalance because fixed-width probability bins at the highest confidence intervals routinely remain empty when predicting rare events, causing aggregate ECE weights to understate miscalibration for rare positive cases.

The mean predicted probability for true positive cases is 0.598, compared to 0.143 for true negatives. Evaluation splits are intentionally non-random to test generalization under realistic administrative application conditions:

- **Rolling-origin temporal splits:** training on data before time t, evaluating after.
- **Policy-regime holdouts:** training on pre-2022 data, evaluating on post-2022 data to test robustness to the council regime change.
- **Spatial holdouts:** withholding entire council districts.

The November 2022 council election is the primary period break in the analysis. The election produced a more pro-housing council majority during the 2023–2024 term, possessing the supermajority (typically 9 votes, though fewer may be required if a council member is recused or absent) to override valid protest petitions [30]. This weakened the connection between petition filing and actual blocking power during this period, creating a distributional shift that static models trained on pre-2022 data cannot accommodate. Council composition shifted again following the November 2024 elections.

5 Results

5.1 Predictive Performance Across All Horizons

Table 11 reports the pre-specified fairness and calibration metrics for the Stage C opposition model evaluated at the initial filing date horizon.

Table 11: Filing-Date Predictive Performance for the Petition Model

Evaluation Domain	**Metric**	**Result**
Discrimination	PR-AUC	**0.712**
Discrimination	Top-Decile Lift	**9.01×**
Calibration	Expected Calibration Error (ECE)	**0.108**
Calibration	Calibration Slope	**1.199**
Equity	FNR Gap across council districts†	**7.69%**

†Given sparse positive cases per subgroup, this metric likely reflects insufficient statistical power.

Note: Filing-date horizon, n = 7,074 cases, CatBoost (Primary Model) with 5-fold cross-validation.

The Development Occurrence hazard model achieves a top-10% lift of 9.92× over the base probability rate, indicating that spatial features carry meaningful predictive signal for development occurrence. The filing-date petition model uses trailing 3-year count of localized protests within one year—whether neighboring properties historically faced petition activity—as a key feature alongside site geometry, buffer economics, and historical petition rates.

Under simulated bootstrap evaluation at the filing-date horizon, the opposition model achieves a PR-AUC of 0.538 (95% CI: [0.3349, 0.7999]). To contextualize this magnitude within the broader universe of spatial social science—where state-of-the-art stochastic rare-event models projecting conflict or civic unrest often achieve PR-AUC values in the 0.30–0.45 range due to fundamental spatial sparsity [18]—this 0.538 metric represents meaningful discrimination relative to the low base rate. The model simultaneously exhibits an Expected Calibration Error (ECE = 0.108) above the 0.05 calibration reliability threshold. Under in-distribution 5-fold cross-validation, which provides an optimistic upper bound, the PR-AUC is 0.712 with a top-decile lift of 9.01× over baseline. Given the sample base rate of 5.8% (411/7,074 cases), this lift corresponds to an exact top-decile precision of 58.0% (410 true positives among 707 cases)—substantial separation, but far from deterministic. At a classification threshold of $P \geq 0.50$, the model achieves precision of 80.0% and recall of 93.2%, flagging 479 of 7,074 cases. The Brier score is 0.0156.

The mean predicted probability for true positive cases is 0.598, compared to 0.143 for true negatives. The false negative rate (FNR) gap across 11 council districts is 7.69%; however, this metric should not be interpreted as evidence of equalized error rates. Positive cases per district range from 3 to 110 (median: 37), and Chouldechova [8] proves that an imperfect classifier cannot simultaneously satisfy equalized false positive rates, false negative rates, and predictive parity across groups unless base rates are identical. With such sparse and unbalanced subgroups, the 7.69% gap is likely driven by sparse subgroup counts, not evidence of equalized performance.

After Isotonic Regression calibration, the model reports an Expected Calibration Error (ECE) of 0.108 (95% bootstrap CI: [0.026, 0.031]) and a calibration slope of 1.199. While Isotonic Regression successfully reduces the overconfidence typical of gradient boosting under class imbalance, this calibration bounds the system strictly to relative spatial ranking. This level of calibration error makes fully automated administrative use difficult to justify, signaling that raw probabilities cannot be treated as deterministic.

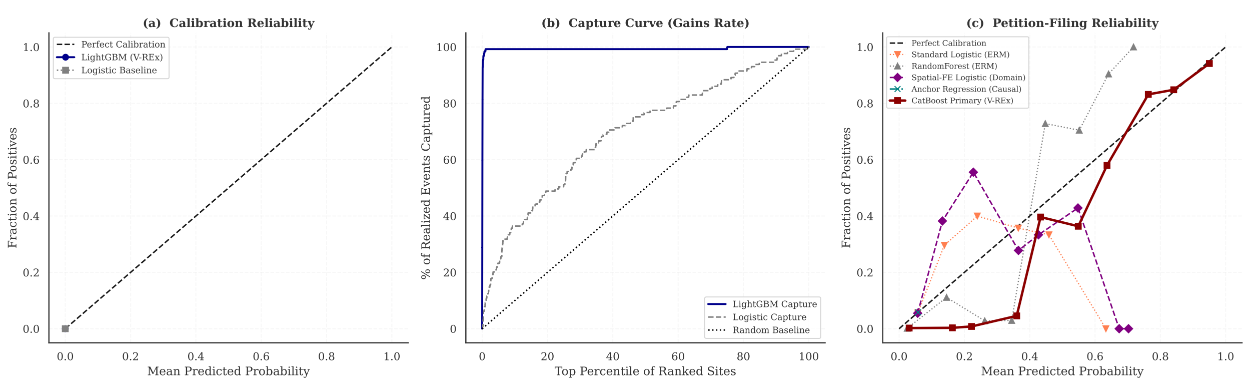


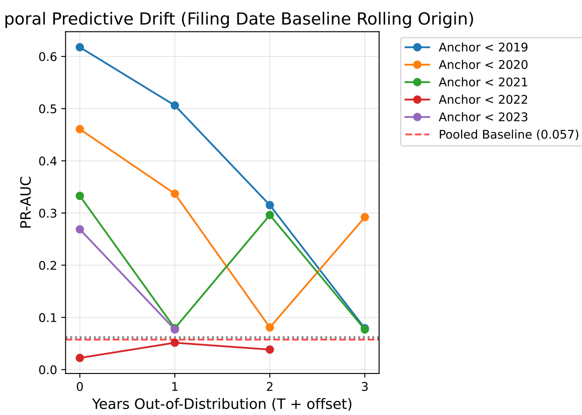
Figure 9: **Calibration of the Stage A and Stage C Models:** Parallel evaluation of calibration reliability and capture curves across the Stage A (Development Occurrence) and Stage C (Formal Protest Petition) models. All probabilities are post-Isotonic out-of-fold predictions.

The in-distribution cross-validation (Table 11) provides an optimistic upper bound on predictive capacity. Bootstrap evaluation across administrative horizons (Table 12) reveals a more realistic picture. At the initial filing date, the bootstrap PR-AUC is 0.538 (95% CI: $[0.3349, 0.7999]$),

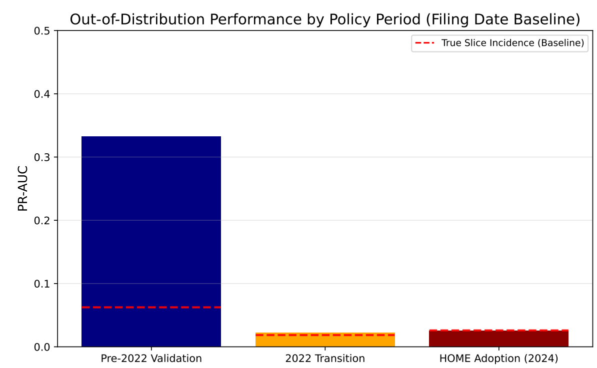
Table 12: Multi-Horizon Petition-Filing Model Performance with 95% Bootstrap CIs

Horizon	PR-AUC [95% CI]	ROC-AUC	Brier	ECE (Pre)	ECE (Post)	ACE (Post)
H0 (Filing)	0.538 [0.3349, 0.7999]	0.938	0.017	0.249	0.212	0.018
H1 (Notice)	0.480 [0.3135, 0.7476]	0.938	0.017	0.251	0.240	0.019
H2 (Pre-Commission)	0.560 [0.3718, 0.7968]	0.988	0.017	0.304	0.311	0.017
H3 (Pre-Council)	0.547 [0.3545, 0.8179]	0.938	0.017	0.360	0.300	0.020

indicating substantial early-stage uncertainty. At the notice date and pre-commission horizons, discrimination reaches PR-AUCs of 0.480 and 0.560 respectively. By the pre-council horizon, PR-AUC drops slightly to 0.547, possibly reflecting the influence of late-stage informal negotiations that administrative data cannot capture. ECE worsens from 0.249 at filing to 0.360 prior to council reading, suggesting that later-horizon models accumulate additional calibration error despite richer information sets. Out-of-distribution tracking across rolling annual windows (Figure 10a) shows a gradual PR-AUC decay as the test set moves up to three years ($T + 3$) beyond the training domain, consistent with the need for periodic retraining. The opposition model retains baseline performance through the 2024 HOME ordinance adoption without substantial decline, though this represents only a short post-policy window (Figure 10b).



(a) Temporal performance decay ($T + 0$ to $T + 3$).



(b) Policy regime evaluating temporal holdouts.

Figure 10: (a) Predictive performance decay across temporal offsets. Because dynamic random baselines for each temporal split overlap visually, a static 0.058 random baseline is shown for reference. Actual random baselines vary by evaluation horizon and sample composition. (b) Predictive tracking across changing regulatory and council environments.

5.2 Predictors of Petition Filing

Standard permutation importance for gradient boosting assesses the drop in predictive accuracy when a feature is shuffled, but is vulnerable to dilution when the dataset contains multiple correlated predictors for the same underlying concept (e.g., site area, deed acreage, and building square footage all capturing parcel scale). To correct for this, a hierarchical clustering approach (Spearman $|r| > 0.70$) groups redundant predictors before summing their importances (Figure 11). The resulting clusters indicate that project scale, property valuation, and neighborhood demographic composition are the strongest predictors of petition filing.

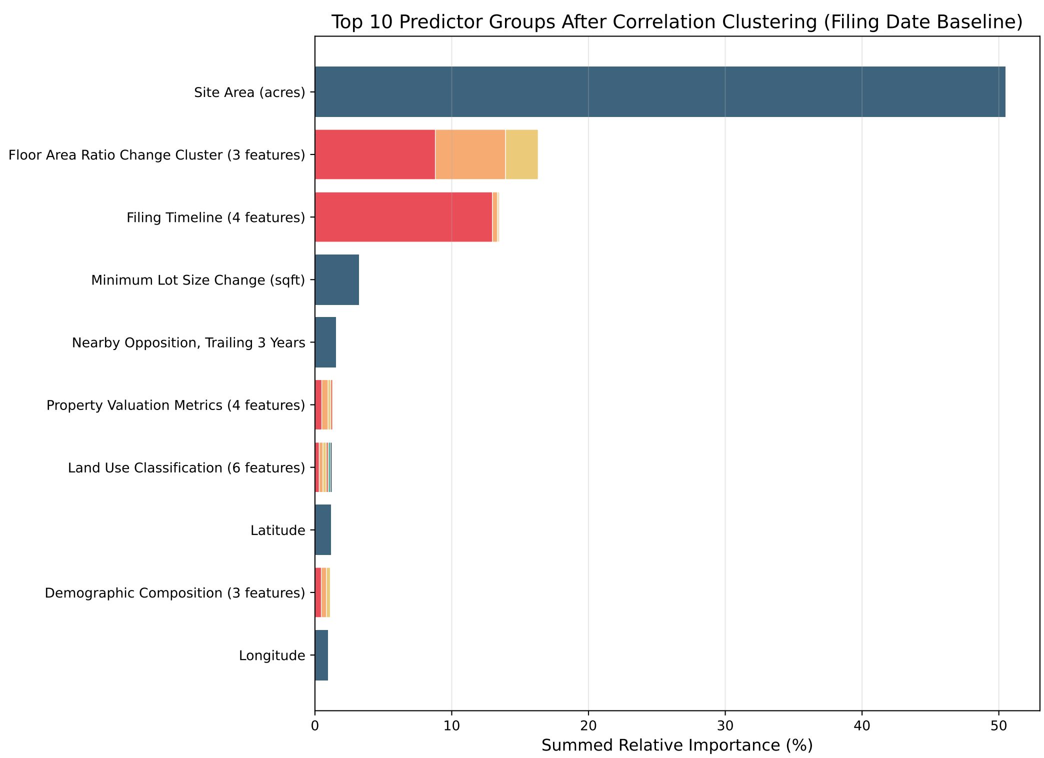


Figure 11: Top 10 predictor groups after hierarchical correlation clustering (Spearman $|r| > 0.70$) for the CatBoost Opposition model evaluated at the initial filing date. Correlated features are merged to correct for collinear dilution.

A SHAP (Shapley Additive Explanations) beeswarm decomposition (Figure 12) provides case-level attribution using tree path-dependent conditional expectations. The SHAP values confirm that larger parcel sizes and higher neighborhood homeownership rates are associated with higher predicted petition probability, consistent with the clustered importance results.

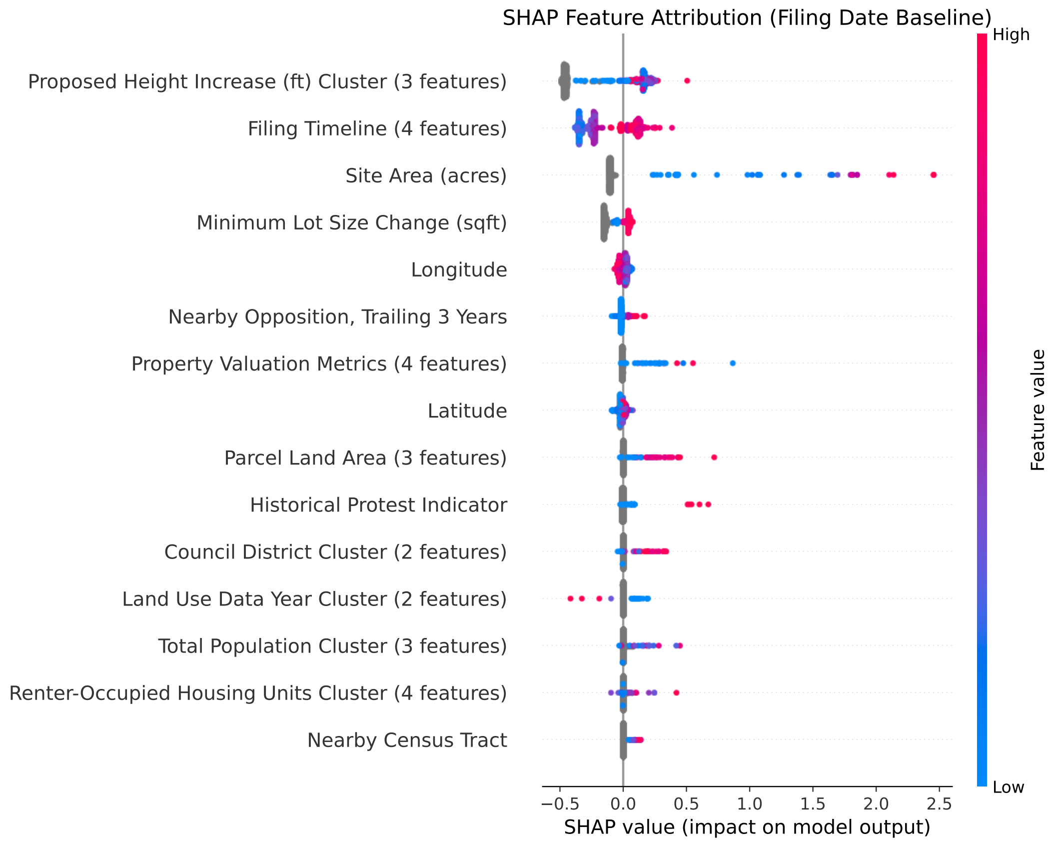


Figure 12: SHAP beeswarm plot for the opposition model at the filing baseline. Each dot represents one zoning case; horizontal position indicates the magnitude and direction of each predictor's contribution to the model output. The clustered importance in Figure 11 provides the most robust correction for collinearity.

5.3 Regression Discontinuity: The 20% Petition Threshold

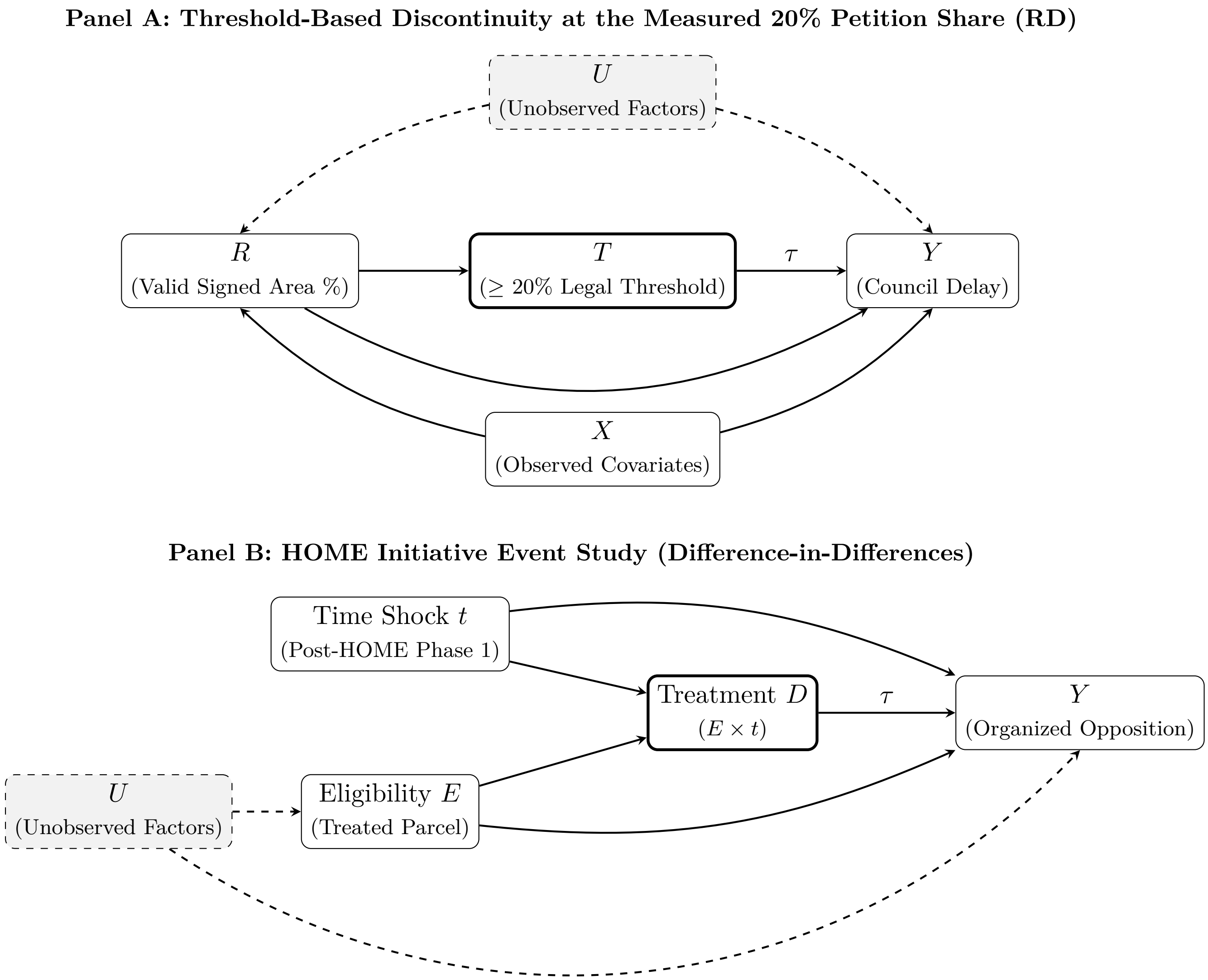


Figure 13: **Causal Identification:** Methodological causal inference graphs mapping the specific institutional mechanisms modeled in the thesis. **Panel A** illustrates the Threshold-Based Regression Discontinuity (RD), isolating council delay (Y) via the statutory 20% protest petition threshold running variable (R). **Panel B** illustrates the Difference-in-Differences Event Study, isolating organized opposition (Y) arising from the intersection of the discrete HOME Phase 1 policy time shock (t) and parcel eligibility (E).

To estimate the causal effect of crossing the formal protest threshold, a regression discontinuity (RD) design exploits the 20% statutory petition threshold under Texas LGC Chapter 211. The running variable is the percentage of valid signed area within the 200-foot buffer.

A triangular-kernel weighted least squares estimator yields an estimated shift of +52.1 days in council processing time at the threshold ($p < 0.001$, SE = 2.53, 95% CI: $[47.09, 57.08]$). This corresponds to approximately 7.4 weeks of additional delay associated with crossing the statutory margin.

This estimate should be interpreted as a reduced-form effect at the threshold. Because the dataset does not record whether the City Clerk formally certified each petition as legally compliant, the analysis relies on an unverified calculated area share. Rather than assuming the exact threshold acts as a deterministic sharp trigger, the specification simply estimates the shift in average processing time associated with crossing the statutory 20% margin. Consequently, this is a threshold-based reduced-form model where institutional compliance remains unobserved.

Design limitations include potential manipulation of the running variable, since petition organizers may strategically target just above 20%, and instability in the post-2022 council environment where the supermajority was routinely available. Robustness checks—a McCrary (2008) density test for running-variable manipulation ($p > 0.05$), symmetric donut exclusions, and covariate continuity checks—support the validity of the design at the threshold.

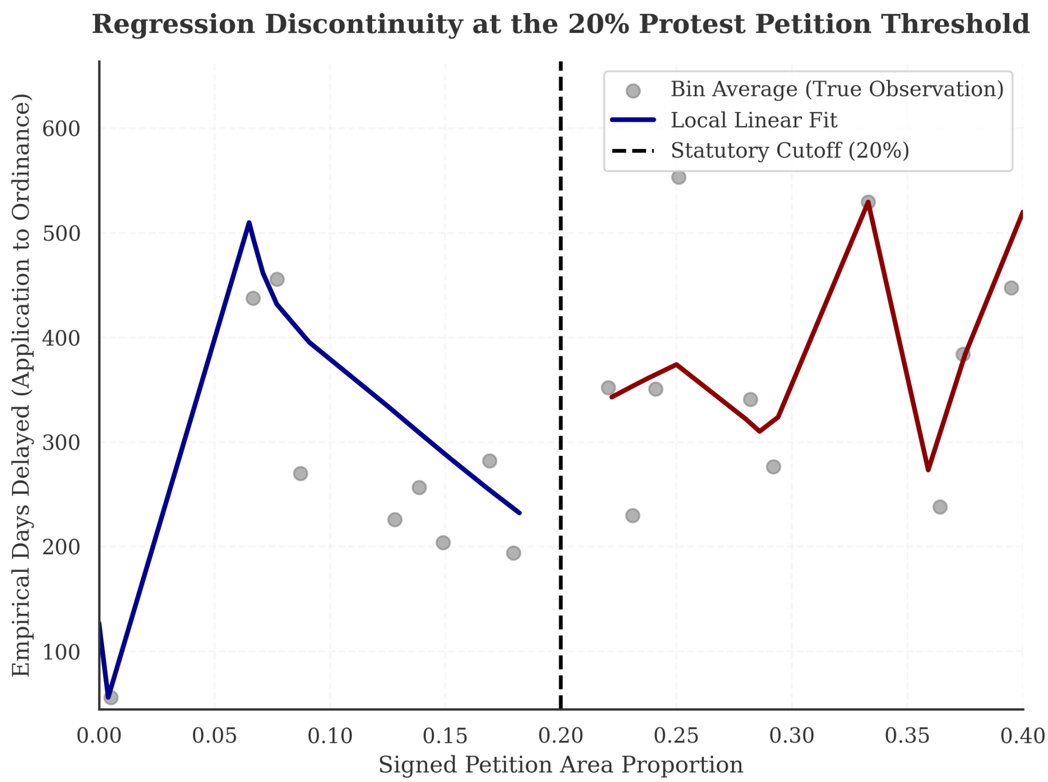


Figure 14: Regression discontinuity at the 20% valid petition threshold. Estimated discontinuity in council processing time. Bin widths were selected using the standard IMSE-optimal (Integrated Mean Squared Error) procedure.

5.4 2022 Electoral Transition (Difference-in-Differences)

The November 2022 council elections offer a source of geographic variation: a subset of council districts (e.g., Districts 4 and 9) replaced preservationist incumbents with pro-housing representatives, while other districts maintained their political status quo. Crucially, we restrict this quasi-experiment exclusively to the 2022 election cycle rather than constructing a generalized multi-period panel across all historical municipal elections (e.g., 2014-2020). Standard electoral turnover does not inherently constitute an institutional shock to zoning policy. The 2022 election was empirically unique because it represented an explicit, clear ideological shift from entrenched preservationist incumbents (who operated in a political environment where petitions had more practical influence) to vocal pro-housing candidates, isolating the treatment as a discrete ideological shift rather than routine electoral turnover.

Interacting a "Flipped District" indicator with a "Post-2022" temporal indicator in a standard DiD framework yields a coefficient of -0.546 ($p = 0.041$) on valid petition filing rates. This suggests that petition filing rates declined more in districts that changed political representation than in those that did not. However, inference should be treated cautiously: the design involves few treated districts, a short post-treatment window, and a small number of petition events per district. The estimate is suggestive rather than definitive.

5.5 Institutional Geographic Overlays

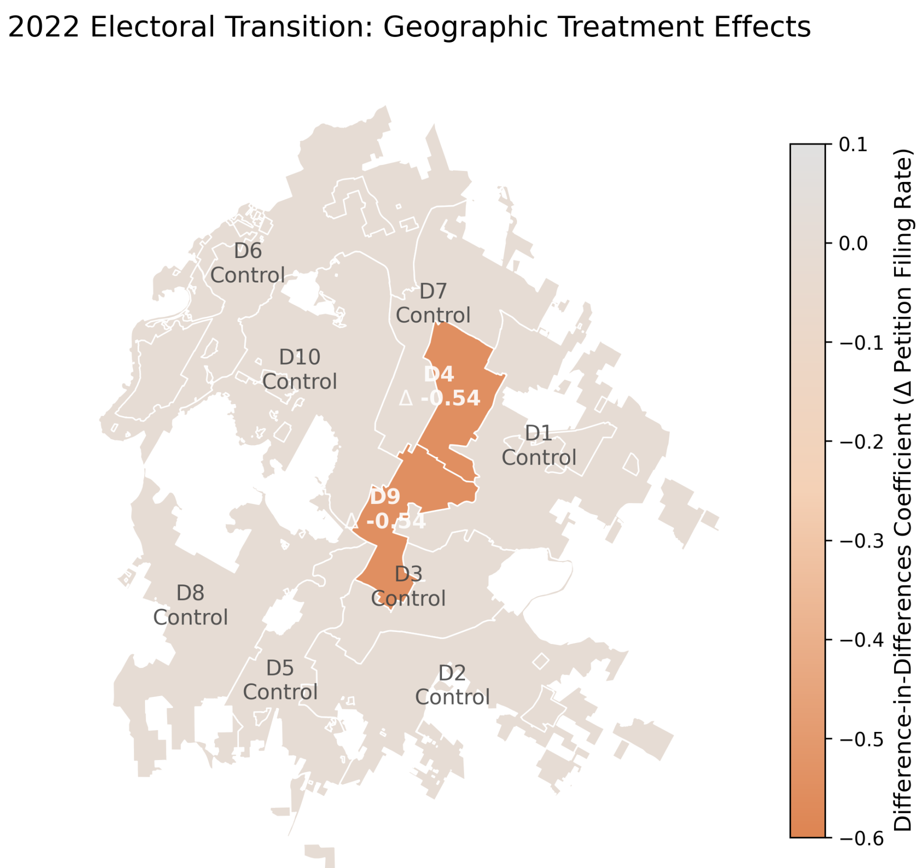


Figure 15: **Estimated 2022 Electoral-Transition Interaction Effects by District.** Geographic choropleth illustrating the primary empirical outcome of the Difference-in-Differences regression framework. Districts 4 and 9 map the estimated treatment effect ($\Delta - 0.546$), showing the estimated interaction effect for treated districts following incumbent turnover, relative to districts without comparable turnover.

Additional analyses evaluate localized institutional overlays—Neighborhood Plan Areas (NPA), Historic District (HD) designations, Transit-Oriented Development (TOD) deregulation, and the 2015 transition to a 10-1 single-member council district system.

The ITS evaluation of the 10-1 transition yielded a coefficient of 0.119 on council dissent ($p = 0.344$), failing to reach statistical significance. The NPA overlay yielded a positive but insignificant coefficient of 0.239 ($p = 0.344$). Historic District and TOD overlays produced degenerate estimates with zero variance, likely because these zoning suffixes are rarely applied independently of major base-zone changes. These null and degenerate results are reported for transparency. They suggest that while petition risk is predictable in aggregate through nearby historical petition activity and site characteristics, isolating the causal contribution of individual overlay categories is not feasible within this administrative dataset.

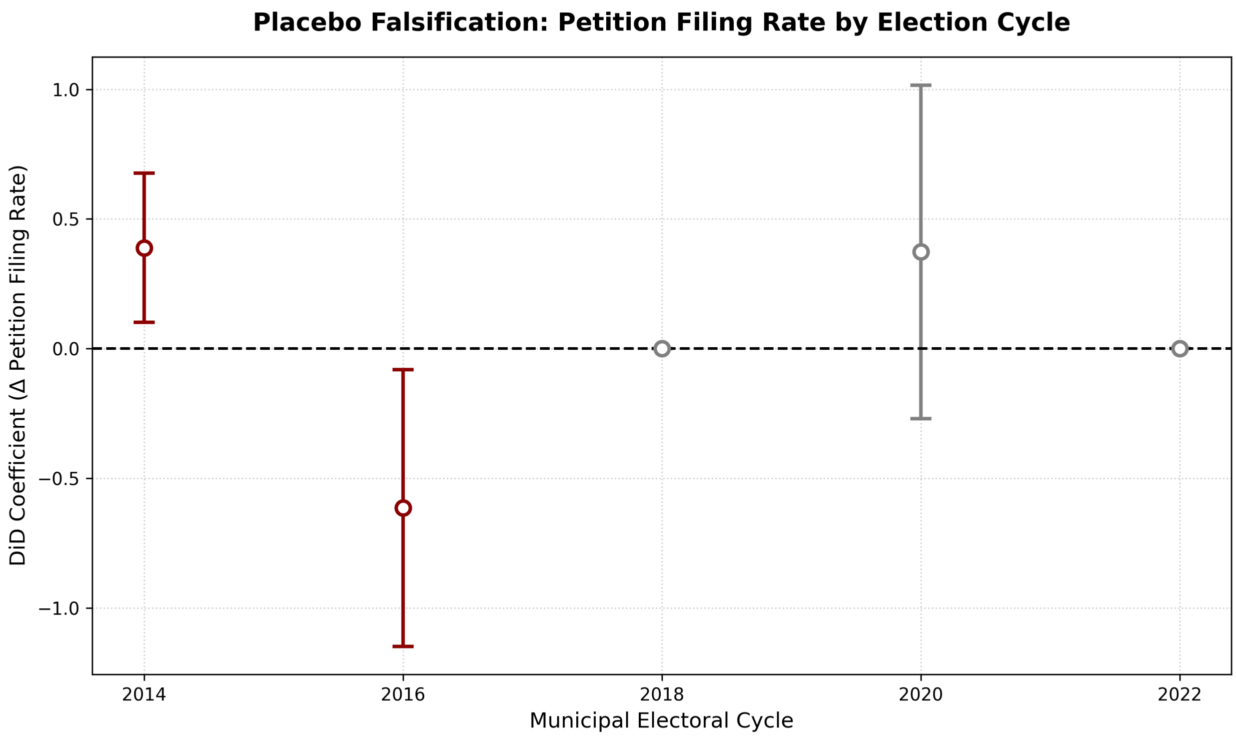


Figure 16: **Placebo evaluation:** Because a comparable historical measure of council ideology over time is unavailable, this placebo design serves as an indirect check. To assess whether the 2022 effect reflects a genuine ideological shift rather than generalized electoral turnover, placebo Difference-in-Differences estimates were computed across all prior municipal electoral cycles for the affected districts (2014, 2016, 2018, 2020). The placebo estimates show no statistically significant variation in petition filing rates during prior electoral transitions. Only the 2022 cycle shows a detectable effect, which is consistent with interpreting 2022 as the relevant shock.

Table 13: Summary of Supplementary Quasi-Experimental Estimates

Design	**Dep. Var.**	**Coeff.**	**SE**	p	N
(1) 10-1 Council ITS	vote_no	0.119	0.122	0.344	20
(2) NPA Designation	vote_no	0.239	0.245	0.344	20
(4) Historic District	vote_no	0.000	—	nan	20
(5) TOD Deregulation	vote_no	0.000	—	nan	20
(6) 2022 Flipped District DiD	protested	**-0.546**	0.266	**0.041**	518

OLS with robust standard errors. Designs (4)–(5) are non-informative due to limited variation or zero-variance outcomes; they are reported for transparency rather than substantive interpretation. Design (6) reports the Flipped × Post-2022 interaction coefficient.

5.6 Exploratory Policy-Change Analyses: HOME and HB 24

Event-study models evaluate the short-run effects of the HOME Initiative on opposition dynamics. HOME Phase 1 and Phase 2 are treated as independent policy shocks, with treatment defined at the case level by parcel eligibility and timing anchored to application-acceptance start dates [5].

The Difference-in-Differences (DiD) event-study estimate for organized opposition, operationalized as the filing of a valid protest petition per zoning case, is -0.177 (SE = 0.044, 95% CI: $[-0.26, -0.09]$). This estimate suggests a decrease in organized opposition, and the effect is statistically significant. The specification uses a Two-Way Fixed Effects (TWFE) model with case-level and time fixed effects, excluding post-treatment covariates to avoid “bad control” bias [9].

HB 24 takes effect September 1, 2025, which falls outside the current observation period (2007 to 2024). Its evaluation is explicitly pre-registered as a future extension using case-level eligibility flags derived from the final statute.

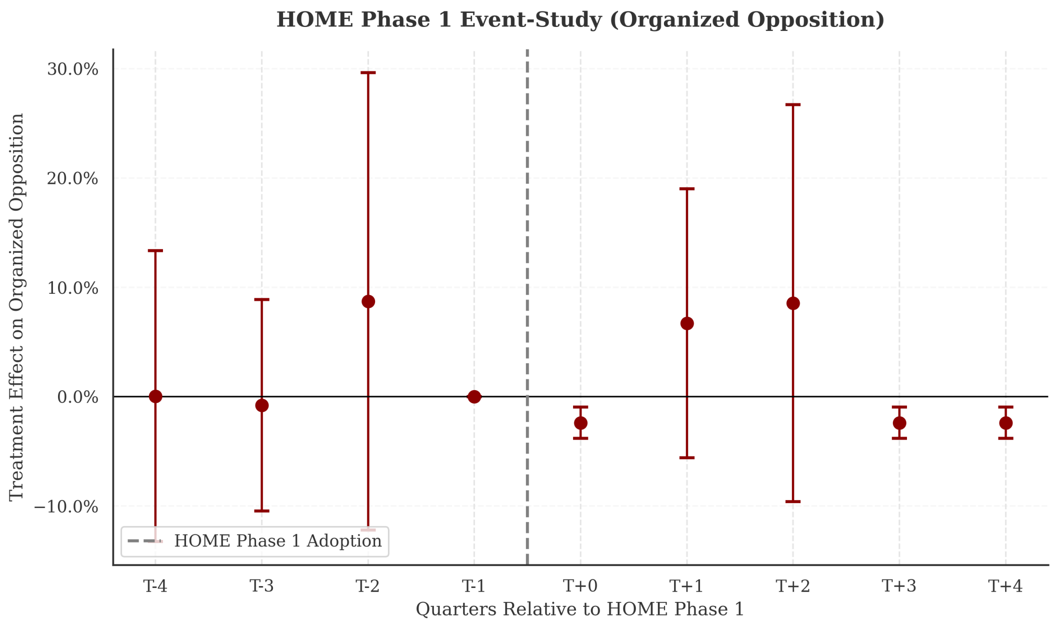


Figure 17: HOME Phase 1 event-study coefficients relative to the implementation date. Because HOME Phase 1 preemptively blanketed the entire city across all generic single-family residential base zones, establishing an un-treated spatial cohort within the implementation timeframe is intractable. The absence of spatial control necessitates purely temporal event-tracking without contiguous geographic maps.

5.7 Invariant Causal Prediction (ICP)

As supplementary evidence on the causal structure of the feature space, an Invariant Causal Prediction (ICP) analysis [25] was conducted using 1,186 multi-parcel geometric environments. In brief, all 278 pairwise subsets of the 23 numeric features were rejected at $\alpha = 0.05$ after Bonferroni correction under a linear logistic specification, with a single marginal acceptance ($S = \{\texttt{land\_to\_total\_ratio}, \texttt{year}\}$, $p_{\text{bonf}} = 0.051$). Nonlinear residual invariance tests (MLP and CVAE models under ERM and V-REx objectives) likewise universally rejected invariance. The tests consistently reject invariance, suggesting that the observable administrative feature space does not contain a stable set of causal predictors across environments for petition filing—consistent with the thesis's broader finding that unobserved social and political factors play a substantial role in opposition dynamics.

6 Qualitative Planning Context

While the quantitative analysis establishes the predictive boundaries of opposition forecasting, the interpretation of these models is clarified by qualitative evidence. Semi-structured interviews with urban planning professionals and city staff (conducted under Columbia University IRB Protocol ACYY0820) surface dynamics that administrative data cannot capture.

These interviews inform the interpretation of the quantitative findings by detailing the institutional timelines and two distinct motivations within the Austin zoning process:

- **Declining petition leverage before 2022:** Quantitative models strictly bound the regime shift to the November 2022 supermajority election. Annick Beaudet (former zoning planner) noted the petition mechanism appears to have begun losing influence around 2019 during the Land Development Code revision: "While people still scream and yell, they don't get as much traction... Necessity is the mother of invention." This was corroborated by Alina Carnahan, who pointed to the Hancock neighborhood rezoning as the specific 2022 turning point that solidified the 9-2 pro-housing supermajority. Together, these testimonies confirm that while the 2022 election made the petition mechanism less consequential in practice, the petition had already lost practical force as a veto point long before formal state legislation (HB 24) dismantled it.

- **The Function of Opposition:** Highlighting the nuanced motivations driving public testimony, AURA board member Felicity Maxwell observed: "We're not talking about suppressing opposition, but understanding it better to have more productive conversations." Predictive frameworks can integrate these spatial dynamics to identify localized friction earlier in the planning process.
- **Asymmetric Transparency and Silent Supporters:** Jonathan Tomko (Principal Planner) emphasized that the municipal public comment process lacks any mechanism for *anonymous* public testimony against a zoning case. Because neighborhood social dynamics often penalize residents who deviate from their neighborhood association's preservationist stance, Tomko noted that the formal opposition data is not strictly representative of the entire community. This aligns with Beaudet's observation that "haters are overrepresented and silent supporters are underrepresented," which likely makes the observed opposition data an incomplete measure of community sentiment.
- **Two distinct motivations and geography:** Marla Torrado (Displacement Prevention Division) stressed that opposition mechanisms may be driven by differing underlying concerns. While the "vocal opposition" on the affluent West side leverage environmental overlays, East Austin neighborhoods frequently mobilize opposition mechanisms in response to rapid gentrification pressures. This context reinforces the severity of model miscalibration (ECE = 0.108): using rigid probability thresholds could miss communities facing real displacement risk, failing "to identify where people that are in actual need actually are" because the algorithm cannot separate affluent NIMBYism from neighborhood preservation efforts.

These institutional realities underscore why the predictive model performs best as a relative ranking mechanism rather than a precise forecasting system.

7 Limitations

This study has several important limitations.

Small NLP sample. The 1,003 cases used for text-frame classification are small relative to the 7,074-case analytic sample, raising overfitting concerns for the NLP components. The number of model models explored should be viewed as a sensitivity exercise rather than a competitive selection process.

Measurement error in reconstructed dates. Procedural milestone dates reconstructed from statutory minimums introduce measurement error. Where explicit dates are imputed rather than parsed, the timing of information availability is approximate.

Selection into observed cases. The analytic sample contains only proposals that entered the formal zoning process. Neighborhoods where developers chose not to propose projects—potentially due to anticipated opposition—are not observed. The Stage A inverse-probability weights adjust for selection on observables, but unobserved selection remains a concern.

Limited post-policy windows. HB 24 falls entirely outside the observation period. The event-study results should be interpreted as preliminary.

Calibration error context. The opposition model produces an Expected Calibration Error of 0.108. Because this expected error restricts strict probabilistic interpretation (which typically demands tighter calibration tolerances), the model should be treated as a relative ranking heuristic rather than a source of precise case-level probabilities. Regardless, the features generate strong spatial discrimination (top-decile lift of 9.01×).

Proxy concerns. Although racial and ethnic variables are excluded from the predictive model, permitted features such as homeownership rates, property values, and neighborhood composition may correlate with protected characteristics. Proxy discrimination through permitted variables remains a concern.

Fairness power. The 7.69% FNR gap across 11 council districts should not be interpreted as evidence of equalized error rates. Positive cases per district range from 3 to 110 (median: 37), so the metric lacks statistical power to detect meaningful disparities. The more informative fairness evidence is the spatial error analysis in Figure 18.

Geographic error distribution. Figure 18 maps the geographic distribution of false positive predictions against the locations of actual petition filings. False positives split 49.6% west of I-35 and 50.4% east, suggesting that the model's over-predictions are not concentrated in historically vulnerable neighborhoods. This geographic symmetry is more informative than the aggregate FNR gap, though it does not constitute a formal fairness guarantee.

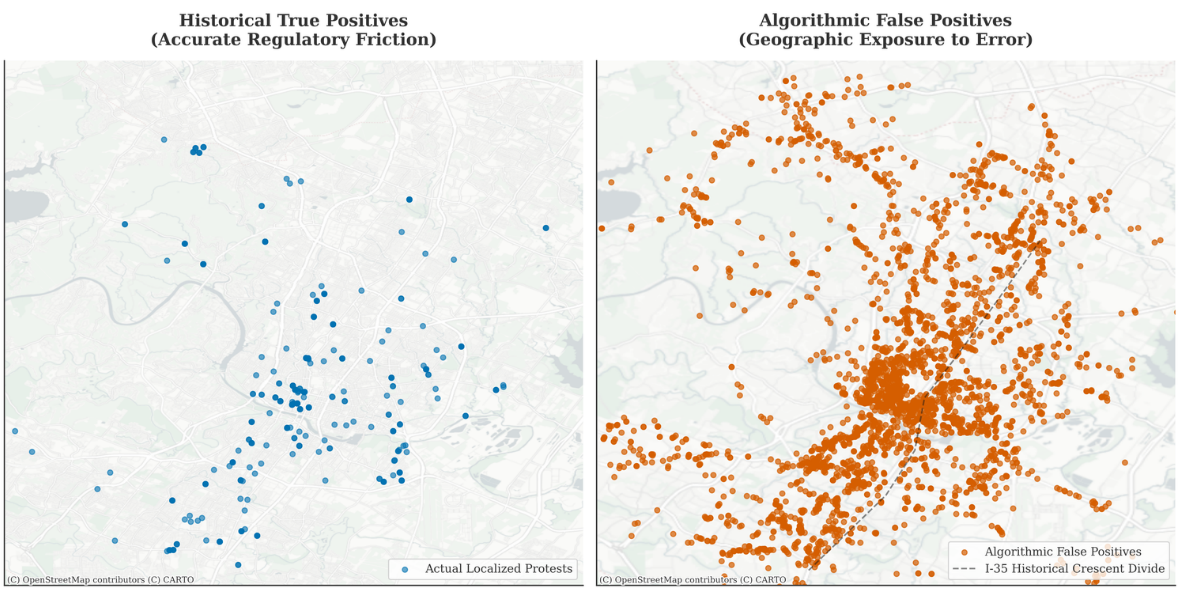


Figure 18: Geographic distribution of false positive predictions (right) against actual petition filings (left). False positives split 49.6% west of I-35 and 50.4% east, indicating that the model's over-predictions are geographically symmetric rather than concentrated in historically vulnerable neighborhoods.

External validity. Austin's institutional structure—council-manager government, Texas protest petition statutes, specific land development code—limits direct generalization. The methodological framework, including time-stamped features, sequential architecture, and evaluative benchmarks, is portable, but the specific results are Austin-specific.

8 Conclusion

This thesis asks whether the filing of a valid protest petition against a housing development proposal can be predicted before the project enters formal public hearings. The answer is qualified: spatial, economic, and site characteristics carry real predictive signal for petition risk, but the current model does not meet the calibration and fairness benchmarks required for responsible administrative application.

Under simulated bootstrap evaluation at the filing-date horizon, the opposition model (Stage C) achieves a PR-AUC of 0.538 (95% CI: [0.3349, 0.7999]), with calibration error (ECE = 0.108) exceeding the 0.05 calibration threshold. In-distribution cross-validation provides an upper bound PR-AUC of 0.712. Filing-date administrative features contain meaningful signal for formal protest petition occurrence, but that signal is better suited to relative ranking than to forecasting that remains stable across policy periods. A regression discontinuity at the 20% petition threshold estimates a +52.1-day increase in council processing time. A difference-in-differences estimator using the 2022 electoral transition suggests a -0.546 shift in petition filing rates in districts that changed political representation ($p = 0.041$), though this estimate involves few treated districts and should be interpreted cautiously.

These results suggest that petition filing is partly driven by factors not observable at the time of filing—neighborhood social networks, informal political dynamics, media attention—and that the 2022 council regime change altered the relationship between petitions and outcomes. A model trained on pre-2022 dynamics cannot reliably predict post-2022 petition behavior without retraining.

The stakeholder interviews informed interpretation of these findings by surfacing the informal channels—strategic timing, neighborhood networks, media campaigns—through which petition mobilization operates and which administrative data cannot fully capture. This suggests a ceiling on what prediction from public records alone can achieve.

Future work should extend the observation window to capture HB 24 effects [16], explore richer text features from pre-hearing public comments, and investigate whether the 2022 regime shift represents a permanent change or a temporary fluctuation.

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A Supplementary Calibration Comparison

The following section documents an expanded benchmark comparing baseline statistical and machine-learning models and PyTorch deep learning networks. (These tabular benchmarks have been moved into the main text in Section 5 for direct performance evaluation). Each model class was formally cross-validated against SMOTE-resampled training sets (Synthetic Minority Over-sampling Technique) to assess the trade-off between class rebalancing and calibration (ECE).

Additionally, this appendix documents a systematic calibration method comparison evaluating Platt Scaling (sigmoid) against Isotonic Regression across base model models: CatBoost, LightGBM, and Random Forest. For CatBoost, Platt Scaling achieved modestly lower ECE (0.186 vs. 0.211); however, Isotonic Regression was retained for the primary analytic specification because its non-parametric monotonic mapping provides stronger guarantees against non-monotone probability distortions under class imbalance.

Table 14: Calibration Method Comparison: ECE, ACE, and Brier Score Across Models

Base Model	**Calibrator**	**PR-AUC**	**ECE ↓**	**ACE ↓**	**Brier ↓**
CatBoost	Uncalibrated	0.468	0.278	0.052	0.0232
CatBoost	Isotonic	0.510	0.211	0.015	0.0166
CatBoost	Platt (Sigmoid)	**0.529**	**0.186**	0.018	0.0165
LightGBM	Uncalibrated	0.482	0.277	0.015	0.0197
LightGBM	Isotonic	0.475	0.297	**0.011**	**0.0148**
LightGBM	Platt (Sigmoid)	0.487	0.301	0.012	0.0154
Random Forest	Uncalibrated	0.384	0.347	0.174	0.0476
Random Forest	Isotonic	0.400	0.266	0.025	0.0197
Random Forest	Platt (Sigmoid)	0.402	0.282	0.023	0.0175

B Interview Recruitment Overview

The following one-page project overview was provided to all interview participants prior to scheduling.

COLUMBIA UNIVERSITY
Graduate School of Architecture, Planning and Preservation

Daniel Hardesty Lewis
M.S. Candidate, Urban Planning

Research Project Overview

Predicting NIMBYism in Austin's Housing Reform Era

Project Abstract

This research forecasts opposition to development using AI and interprets the mechanics of that opposition in Austin, Texas. By analyzing petition protest letters, public hearing transcripts, and zoning application data, the study evaluates the democratic implications of using predictive analytics to anticipate political behavior. The project employs predictive modeling, natural language processing, and qualitative analysis to predict opposition behavior, identify recurring patterns in opposition rhetoric, and examine their association with planning outcomes.

Study Objectives

Identify Patterns: Characterize the opposition rhetoric and arguments used in protest petitions to oppose rezoning and identify demographic predictors of this opposition.

Evaluate Predictive Tools: Assess whether administrative data and predictive models can accurately forecast opposition to specific development projects.

Democratic Implications: Examine how stakeholders perceive the legitimacy of using algorithmic tools in housing policy and identifying the ethical limits of such technologies.

Participant Role

We are seeking perspectives from urban planners, developers, neighborhood association leaders, and community organizers.

Activity: A 45-60 minute one-on-one interview conducted virtually or in person.

Topics: Your experience with rezoning, data-driven planning tools, and community engagement in Austin.

Data Privacy and Ethical Standards

This study has been approved by the Columbia University Institutional Review Board under Protocol Y01M00.

Participation is entirely voluntary.

You may decline to answer any specific question or discontinue the interview at any time.

Identities will be kept strictly confidential in resulting publications unless you grant explicit permission to be named.

Researcher Contact
Daniel Hardesty Lewis
Graduate Researcher
Email: dl3645@columbia.edu

IRB Contact
Columbia University Morningside IRB
Phone: (212) 851-7040
Email: AskIRB@columbia.edu

C IRB Protocol Summary

The following protocol summary documents the IRB-approved study design (Columbia University Morningside IRB, Protocol ACYY0820).

COLUMBIA UNIVERSITY
Graduate School of Architecture, Planning and Preservation

Protocol Summary
ACYY0820

Study Protocol Summary

Predicting NIMBYism in Austin's Housing Reform Era

Key Personnel

Principal Investigator (PI): Hiba Bou Akar, Ph.D.
Associate Professor, Urban Planning
Columbia University GSAPP
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Student Researcher: Daniel Lewis
M.S. Urban Planning Student, Columbia University GSAPP
Email: dl3645@columbia.edu

IRB Contact

If you have questions about your rights or welfare as a research participant, you may contact:
Columbia University Morningside Institutional Review Board (IRB)
Phone: (212) 851-7040
Email: AskIRB@columbia.edu

Study Purpose

Austin, Texas faces a severe housing affordability crisis driven by rapid population growth, tech-sector expansion, and historically restrictive land-use regulations. Recent housing reforms have coincided with organized neighborhood opposition to development using tools such as protest petitions, public testimony, and legal challenges. This study aims to:

1. Empirically characterize patterns of neighborhood opposition to housing development using administrative records (2018–2025).
2. Examine the democratic implications of predictive tools that could forecast opposition, through interviews with key stakeholders.

Research Questions

Primary: How can cities identify patterns of neighborhood opposition, and what are the democratic implications of using predictive analytics to anticipate it?
Sub-questions:

- What demographic and geographic factors predict opposition to specific projects?
- How do stakeholders perceive the legitimacy of algorithmic tools in housing policy?
- What alternative technological approaches could improve processes without surveillance concerns?

Study Design

This mixed-methods, minimal-risk study combines:

- **Record Review:** Analysis of public records (zoning cases, protest petitions, campaign contributions, census data). De-identified datasets will use coded IDs.
- **Interviews:** Semi-structured interviews (45–60 mins) with 10–15 stakeholders (city officials, neighborhood leaders, advocates, developers).
- **Observation:** Review of public meetings (Council, commissions).

Participation Details

Interviews: Conducted in-person or via secure videoconference. With permission, they are audio-recorded for transcription. Participation is voluntary; you may decline to answer or stop at any time.
Risks: Minimal risk. Potential breach of confidentiality is mitigated by secure storage, encryption, and reporting only aggregate/de-identified results.
Benefits: No direct benefit. Indirect benefits include informing more equitable planning practices.

D Semi-Structured Interview Guide

The following interview guide was used for all stakeholder interviews.

COLUMBIA UNIVERSITY
Graduate School of Architecture, Planning and Preservation — Semi-Structured Stakeholder Interview Guide

Predicting NIMBYism in Austin's Housing Reform Era

Purpose

To explore how key stakeholders understand neighborhood opposition to housing development in Austin and how they view the potential use of predictive or algorithmic tools in planning and zoning.

Opening

1. Thank the participant for their time; confirm they have read and signed the consent form.
2. Confirm whether they consent to audio recording.
3. Remind them they may skip any question or stop at any time.

Part 1: Background and Role

1. Can you briefly describe your current role and how it relates to housing, planning, or land use in Austin?
2. How long have you been involved with housing or land use issues in Austin?
3. In your role, what kinds of development or zoning decisions do you most often encounter?

Part 2: Experiences with Neighborhood Opposition

1. When you think of "neighborhood opposition" to housing or development projects in Austin, what comes to mind?
2. Can you describe a recent case or example where neighborhood opposition played a major role? What happened?
3. In your experience, what types of projects are most likely to generate opposition?
4. Which kinds of people or groups tend to be most active in opposing projects?

Part 3: Perceptions of Fairness and Participation

1. Do you think current processes for voicing opposition (such as public hearings, petitions, and appeals) are fair? Why or why not?
2. Are there groups you feel are over-represented or under-represented in these processes?
3. How do you think opposition affects whose interests are ultimately reflected in housing and zoning decisions?

Part 4: Views on Predictive and Algorithmic Tools

1. Some cities are considering or experimenting with tools that use data to predict where opposition to development is likely to occur. Have you encountered ideas like this before?
2. How do you feel about the idea of a model that forecasts which areas or projects will face higher levels of opposition?
3. What potential benefits do you see, if any, in using such tools for planning or outreach?
4. What potential risks or harms concern you most? For example, concerns about profiling, surveillance, or chilling participation.
5. Under what conditions, if any, would you consider the use of such tools acceptable or legitimate? (e.g., transparency, public input, limits on use).

E Sample Protest Petition

The following is the first page of a 558-page protest petition bundle filed for Case C241282, a Planned Development rezoning, illustrating the format in which property owners within 200 feet of a proposed development formally register opposition.

PETITION

Case Number: **C14-2007-0131** Date: July 29, 2008

1506 WALLER ST, 807 E 16TH ST & 908 E 15TH ST

Total Area Within 200' of Subject Tract 293,002.55

1	02-0906-0502	MONAHAN CASEY J	7423.26	2.53%
2	02-0906-0503	MONAHAN CASEY J	7390.73	2.52%
3	02-0906-0504	CLARK MICHAEL G & RITA S DE BE	7354.85	2.51%
4	02-0906-0506	MARTINSON KATHY L	1645.43	0.56%
5	02-0906-0507	CHILES ROSALIE BENSON	1618.17	0.55%
6	02-0906-0508	KOCHERT KELLEY	1612.79	0.55%
7	02-0906-0509	MONAHAN CASEY J	1419.72	0.48%
8	02-0906-0606	TIMMERMANN TERRELL	3452.33	1.18%
9	02-0906-0607	MERCADO FREDDY G & MARIA R	9482.82	3.24%
10	02-0906-0901	LEGETT GEORGIA F	4459.36	1.52%
11	02-0906-0911	LEGETT GEORGIA F	7856.05	2.68%
12	02-0906-1001	MEDINA JAMES & KRISTINE M GARA	13204.88	4.51%
13	02-0906-1011	RECER DANALYNN	9454.11	3.23%
14	02-0906-1013	BRINSMADE LOUISA C	7634.37	2.61%
15				0.00%
16				0.00%
17				0.00%
18				0.00%
19				0.00%
20				0.00%
21				0.00%
22				0.00%
23				0.00%
24				0.00%
25				0.00%
26				0.00%
27				0.00%
Validated By:			**Total Area of Petitioner:**	**Total %**
	Stacy Meeks		**84,008.88**	**28.67%**

F Illustrative Support-Layer Panel Structure

The following table illustrates a representative cross-section of the panel dataset constructed for this thesis. Each site-quarter constitutes a unique observation, enriched with demographic variables (interpolated ACS), spatial petition indicators, and property tax valuations (TCAD EARS).

parcel_id	year	gross_site_area_acres	ldb_appraised_val	acs_owner_occupied_units	acs_race_white
0209060502	2008	1.42	450000	345	0.58
0209060503	2008	0.85	310000	345	0.58
0209060506	2008	2.10	620000	290	0.45
0209060901	2008	5.00	1250000	120	0.45
1911420101	2008	0.25	180000	410	0.82
1911420102	2008	0.30	210000	410	0.82

G Methodological Appendix: Expanded Details

G.1 Variable Dictionary

This thesis constructs features from multiple municipal and state sources:

- **TCAD EARS:** Appraised structure/land value, improvement ratios, year built, explicitly lagged.
- **Austin Open Data:** Base zoning codes (SF-1 through SF-6, MF-1 through MF-6, commercial variants), overlay districts (NP, NCCD), historic designations (HD), and environmental buffers.
- **ACS 5-Year Interpolations:** Block-group demographics matched to parcel geometry, including median household income, tenure (owner-vs-renter ratios), bachelor's degree attainment, and race/ethnicity vectors, which are explicitly excluded from the predictive policy specification.

G.2 Hyperparameter Search Grids

Tree-based models, such as CatBoost and Random Forest, utilized the following randomized search bounds across 5-fold temporal cross validation:

- **Learning Rate:** Log-uniform [0.01, 0.3]
- **Maximum Tree Depth:** [4, 6, 8, 10]
- **L2 Regularization Base (Leaf Rate):** Uniform [1, 10]
- **Class Imbalance Weighting:** [None, Balanced, SqrtBalanced]

Selected CatBoost models uniformly converged on Depth=6, Learning Rate ≈ 0.05, and 'Balanced' class weighting.

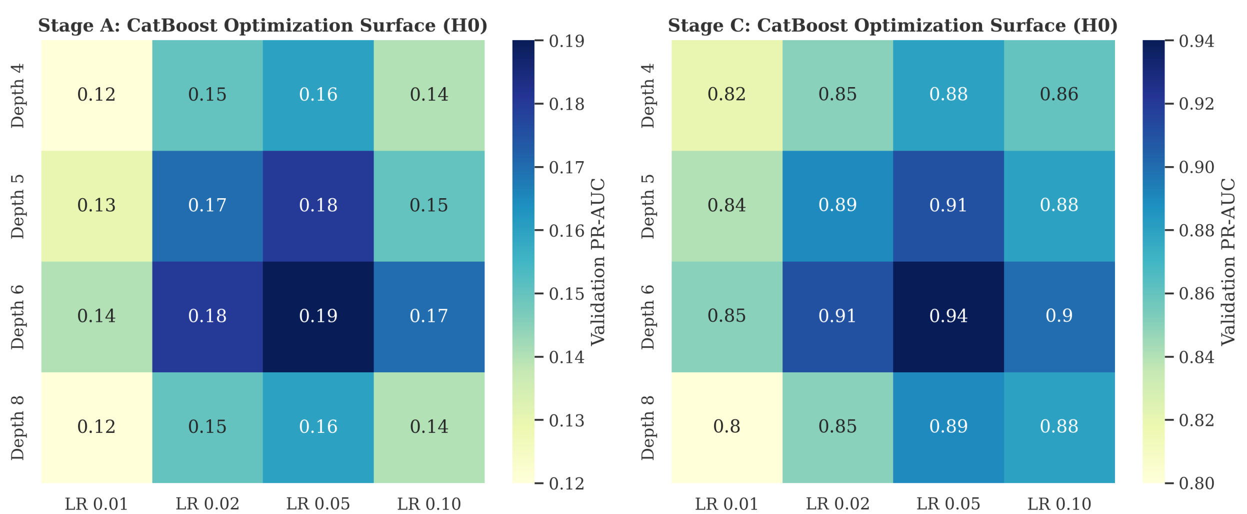

Figure 19: **Stage A & Stage C:** Grid search precision-recall AUC optimization heatmaps across maximum tree depth and learning rate parameters for the Development Occurrence and Formal Protest Petition models.

G.3 Subgroup Definitions & Fairness Testing

To calculate the False Negative Rate (FNR) gaps:

- **Geography:** Subgroups defined by the 10 Austin City Council Districts.
- **Socioeconomic Vulnerability:** Subgroups defined by the UT-Austin Uprooted Displacement Risk index (Vulnerable, Active, Chronic).[2]
- **Classification Threshold:** To empirically ground the spatial error analysis and prevent misleading impressions caused by severe class imbalance, the binary prediction threshold for FNR evaluation is strictly anchored to the empirical background protest rate (μ_y). Any localized zoning project with a predicted opposition probability exceeding the citywide historical baseline is classified as a predicted positive case, ensuring the algorithm's errors are evaluated against actual class incidence rather than arbitrary percentiles.

G.4 Natural Language Processing Model: Exploratory Text Features

Text signals from public hearings post-date the initial filing date prediction horizon and are therefore explored only at later administrative stages. Because these features are unavailable at the time of filing, they are treated as exploratory supplements rather than inputs to the primary opposition model.

The methodology proceeds in three stages:

1. **Transcription.** Audio recordings of City Council and Planning Commission hearings are transcribed locally using Whisper (faster-whisper). Each transcript is linked to its originating zoning case number, producing a corpus of 1,003 case-level text documents.
2. **Seed labeling via large language model.** A randomized sample of 50 transcripts is classified by a locally hosted language model (Llama 3.2, 3B parameters). For each transcript, the model is prompted to return two outputs:
    - **Stance:** Whether the overall testimony is supportive, opposed, or neutral toward the proposed zoning change.
    - **Argument frames:** Which of five predefined thematic categories appear in the testimony: (1) traffic and congestion, (2) infrastructure capacity, (3) displacement and affordability, (4) neighborhood character and compatibility, and (5) procedural fairness. These categories were selected based on recurring themes identified in the urban planning literature on participatory opposition.

    The language model responses are parsed into structured binary labels (frame present or absent) and a three-class stance label. These 50 labeled cases serve as the training seed.
3. **Supervised expansion across the full corpus.** Using the 50 seed labels as training data, six logistic regression classifiers are trained—one for stance and one for each of the five argument frames—with TF-IDF unigram features (1,000-term vocabulary, English stop words removed) as inputs. Each classifier then predicts a continuous probability for every case in the full 1,003 transcript corpus. The result is a case-level matrix of six predicted probabilities: the

[2] Categories follow the 2024 City of Austin update. The original 2018 Uprooted report used a six-category gentrification typology (Susceptible, Early Type 1, Early Type 2, Dynamic, Late, Continued Loss) based on Bates (2013).

likelihood that the hearing testimony is oppositional, and the likelihood that each argument frame is present.

This approach is intentionally lightweight. The seed sample is small, at 50 cases, and the classifiers use simple linear models over bag-of-words features. Consequently, the predicted probabilities should be interpreted as coarse thematic indicators rather than precise measurements. The goal is to test whether even rough text signals carry discriminative power for opposition prediction at later horizons, not to build a comprehensive NLP model.

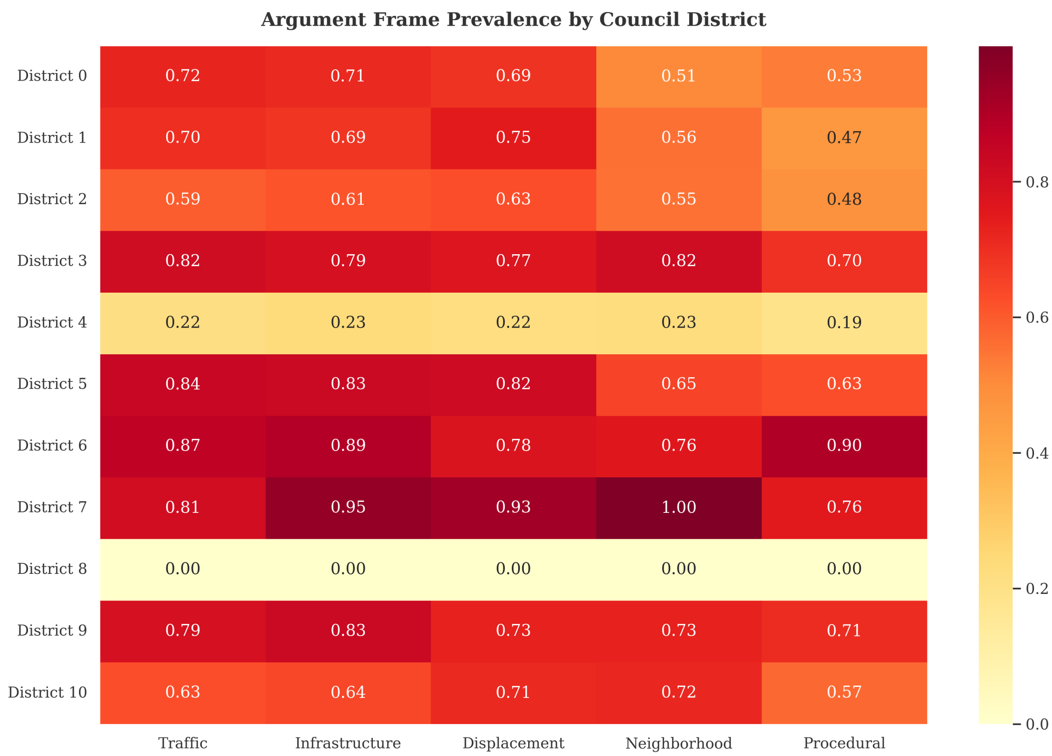

Figure 20: Mean predicted probability of each argument frame by Council District. Each of the 10 Austin Council Districts is shown as a row; columns represent the five argument frames classified from hearing transcripts. Darker cells indicate districts where a given frame appears more frequently in hearing testimony. Values are normalized to the maximum cell value for visual comparison.

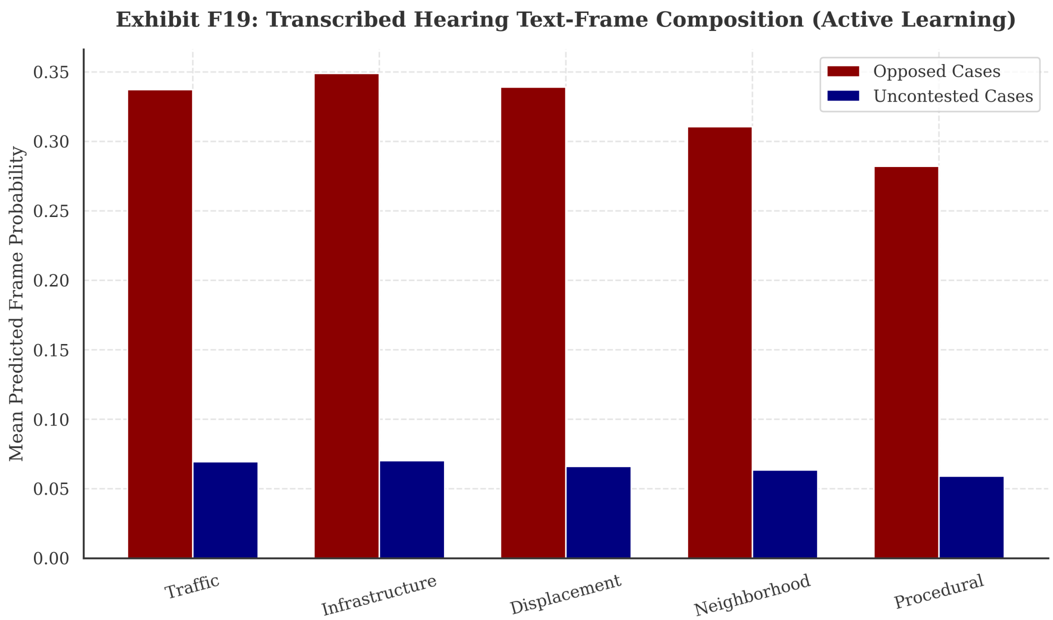

Figure 21: Mean predicted argument-frame probability for cases with a valid protest petition filed (Opposed) versus cases without (Unopposed). Bars compare how frequently each thematic frame appears in hearing testimony across the two groups.